

Strava Fitness Data Analytics Case-Study

Bellabeat: How Can a Wellness Technology Company Play It Smart?

From recorded days to credible habits

Portfolio case study | Supplied Fitbit / MTurk data | 12 April–12 May 2016

The deck title is retained for submission continuity. These are **Fitbit exports, not Strava product telemetry**.

Executive decision: Focus on the **Bellabeat App/Membership**, interpreted as the product option in the case brief. Activity and sleep onboarding, reminder timing and retention experiments can be tested together in an app. These exports contain no hydration, purchase, acquisition-cost or subscription-conversion data, so they cannot justify a Spring launch, hardware revenue forecast or an ROI claim. Recommendations below are proposed experiments, not claims about current product features.

Objective: Describe device-use patterns, separate recording quality from movement, and propose measurable marketing experiments for this one product.

Stakeholders: Urška Sršen (co-founder and CCO), Sando Mur (co-founder), and Bellabeat marketing analytics team.

Business questions:

1. How much of the observed activity is credible under a transparent wear-quality proxy?
2. Which features have broad versus concentrated adoption?
3. When do activity patterns create opportunities for opt-in reminders?
4. How do sleep, activity and exertion relate, with repeated users handled correctly?
5. Which behavioural segments could guide onboarding and retention experiments?

Tools: Python, pandas, NumPy, SciPy, scikit-learn, Matplotlib, Seaborn, PyArrow and SQLite.

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0. Configuration

Takeaway: All paths are relative; both analytical paths rebuild from the supplied CSVs.

Run from the `outputs/` directory. Put the extracted original archive under `../raw/`, or set `FITBIT_RAW_DIR`. The database is regenerated if absent. The first full run imports all 18 files and can take several minutes. Parquet aggregates are refreshed for downstream use. No Power BI or Streamlit application is created.

```
from pathlib import Path
import os
import sqlite3
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from IPython.display import display
from load_csv_to_sqlite import import_files
from run_sql import run_sql
from analysis_utils import prepare_data, plot_chart, PALETTE, SEED

BASE = Path.cwd()
RAW_DIR = Path(os.environ.get("FITBIT_RAW_DIR", str(BASE.parent / "raw")))
np.random.seed(SEED)
sns.set_theme(style="whitegrid", palette=PALETTE)
pd.set_option("display.max_columns", None)
pd.set_option("display.max_colwidth", None)
pd.set_option("display.max_rows", 35)
for folder in ["processed", "charts", "sql_outputs"]:
    (BASE / folder).mkdir(exist_ok=True)
if not (BASE / "bellabeat.db").exists():
    import_files(RAW_DIR, BASE / "bellabeat.db")
execution_log = run_sql(BASE)
assert execution_log.status.eq("PASS").all()
print(f"SQL validation: {len(execution_log)} statements executed successfully")
data = prepare_data(BASE)
d, s, w, u = [data[key] for key in ["d", "s", "w", "u"]]
```

schema PASS 67
inventory PASS 18
missing_invalid PASS 259
activity_quality PASS 1
overlap PASS 3
drop_clean_daily_activity PASS 0
drop_clean_sleep PASS 0
drop_clean_weight PASS 0
drop_hourly_activity PASS 0
drop_user_summary PASS 0
drop_weekly_activity PASS 0
create_daily PASS 0
create_sleep PASS 0
create_weight PASS 0
create_hourly PASS 0
create_users PASS 0
create_weekly PASS 0
drop_hr_hourly PASS 0
drop_hr_bands PASS 0
drop_met_hourly PASS 0
drop_minute_sleep_logs PASS 0
create_hr_hours PASS 0
hr_index PASS 0
create_hr_bands PASS 0
create_mets PASS 0
create_sleep_logs PASS 0
descriptive PASS 1
step_distribution PASS 5
user_quartiles PASS 33
user_segments PASS 3
weekday_weekend PASS 2
hourly_peaks PASS 24
sleep_activity PASS 410
calories_intensity PASS 5
engagement PASS 33
weekly_retention PASS 5
who_guideline PASS 4
heart_rate_zones PASS 4
weight_behavior PASS 8
hr_intensity PASS 5990
user_rankings PASS 33
rolling_activity PASS 846
multimodal_daily PASS 846
correlation PASS 1
insight_user_segments PASS 3
insight_weekday_weekend PASS 2
insight_hourly_peaks PASS 24
insight_sleep_activity PASS 410
insight_calories_intensity PASS 5
insight_engagement PASS 33
insight_weekly_retention PASS 5
insight_who_guideline PASS 4
insight_heart_rate_zones PASS 4
insight_weight_behavior PASS 8
reconciliation PASS 1

```

hr_sampling_audit PASS 723
hr_value_audit PASS 1
minute_code_audit PASS 3
SQL validation: 58 statements executed successfully
Deduplicate activity: 940 rows x 15 -> 940 rows x 15
Apply wear-quality proxy: 940 rows x 17 -> 846 rows x 17
Engineer daily features: 846 rows x 17 -> 846 rows x 26
Deduplicate sleep: 413 rows x 5 -> 410 rows x 5
Validate sleep: 410 rows x 5 -> 410 rows x 5
Validate weight; retain missing Fat: 67 rows x 8 -> 67 rows x 8
Restrict hours to retained activity dates: 22,099 rows x 3 -> 20,045 rows x 11
Aggregate unique sleep minutes to episodes: 188,521 rows x 4 -> 459 rows x 8
Aggregate user features and attach personas: 846 rows x 27 -> 33 rows x 17
Join sleep to exact adjacent activity dates: 410 rows x 11 -> 410 rows x 6
Aggregate complete study windows: 846 rows x 27 -> 125 rows x 7
Join hourly HR, intensity and raw METs: 6,013 rows x 6 -> 5,990 rows x 9

```

1. Data sources and data dictionary

Takeaway: 18 files contain overlapping grains; none provides demographics or app revenue.

Source supplied in the brief: Kaggle/Mobius Fitbit public export collected through Amazon Mechanical Turk. The deck cites DOI [10.5281/zenodo.53894](https://doi.org/10.5281/zenodo.53894) . Provenance and consent are reported by the brief, not independently certified here. The deck's broader March–May collection description does not describe this archive's daily analysis window.

The table below is based on a full scan, not a sample. Date bounds are raw export bounds, so minute sleep can start the previous evening and wide minute files can extend beyond the selected daily window. `source_inventory` includes source SHA-256 hashes and file sizes; `column_audit` lists **every column**, SQLite type, missing count and negative count.

table_name	grain	rows	columns	users	start	end	purpose
dailyActivity	user-day	940	15	33	2016-04-12	2016-05-12	Canonical activity and calories
dailyCalories	user-day	940	3	33	2016-04-12	2016-05-12	Redundant daily calories
dailyIntensities	user-day	940	10	33	2016-04-12	2016-05-12	Redundant daily activity minutes/distances
dailySteps	user-day	940	3	33	2016-04-12	2016-05-12	Redundant daily steps
heartrate_seconds	user-timestamp	2483658	3	14	2016-04-12 00:00:00	2016-05-12 16:20:00	Irregular heart-rate observations
hourlyCalories	user-hour	22099	3	33	2016-04-12 00:00:00	2016-05-12 15:00:00	Hourly calorie rhythm
hourlyIntensities	user-hour	22099	4	33	2016-04-12 00:00:00	2016-05-12 15:00:00	Hourly raw intensity score
hourlySteps	user-hour	22099	3	33	2016-04-12 00:00:00	2016-05-12 15:00:00	Hourly movement timing
minuteCaloriesNarrow	user-timestamp	1325580	3	33	2016-04-12 00:00:00	2016-05-12 15:59:00	Minute energy estimates
minuteCaloriesWide	user-hour with 60 minute columns	21645	62	33	2016-04-13 00:00:00	2016-05-13 08:00:00	Wide minute energy; different coverage
minuteIntensitiesNarrow	user-timestamp	1325580	3	33	2016-04-12 00:00:00	2016-05-12 15:59:00	Minute intensity codes
minuteIntensitiesWide	user-hour with 60 minute columns	21645	62	33	2016-04-13 00:00:00	2016-05-13 08:00:00	Wide minute codes; different coverage

table_name	grain	rows	columns	users	start	end	purpose
minuteMETsNarrow	user-timestamp	1325580	3	33	2016-04-12 00:00:00	2016-05-12 15:59:00	Raw METs exertion field
minuteSleep	user-timestamp-log	188521	4	24	2016-04-11 20:48:00	2016-05-12 09:56:00	Sleep episode/code timestamps
minuteStepsNarrow	user-timestamp	1325580	3	33	2016-04-12 00:00:00	2016-05-12 15:59:00	Minute movement
minuteStepsWide	user-hour with 60 minute columns	21645	62	33	2016-04-13 00:00:00	2016-05-13 08:00:00	Wide minute movement; different coverage
sleepDay	user-day	413	5	24	2016-04-12 00:00:00	2016-05-12 00:00:00	Daily sleep totals
weightLogInfo	user-timestamp	67	8	8	2016-04-12 06:47:11	2016-05-12 23:59:59	Weight/BMI logging behaviour

Field family	Meaning and analytical treatment
Id, logId / LogId	64-bit identifiers, never numeric predictors
ActivityDate/Day/Hour/Minute, SleepDay, Date, Time, date	US dates explicitly parsed at import; ISO text in SQLite, explicit ISO parsing in pandas
TotalSteps / StepTotal / Steps00–59	Counts of recorded steps at the stated grain
Calories / Calories00–59	Exported energy estimate, treated as kcal; daily totals include basal energy and are not exercise calories
Very/Fairly/LightlyActiveMinutes; SedentaryMinutes	Device-classified minutes; not verified on-body wear duration
TotalDistance, TrackerDistance, activity distances	Export units undocumented in attached files; retained without km/mile interpretation
Intensity, TotalIntensity, AverageIntensity	Raw device score/code, not a physical unit or minutes of exercise
METs	Raw integer export score; scaling not verified, so no division by 10 or clinical MET threshold
Heart-rate Value	Recorded beats/minute; descriptive bands, no age-based zones
TotalMinutesAsleep, TotalTimeInBed, TotalSleepRecords	Daily sleep minutes, in-bed minutes and record count
minuteSleep value	Unverified state code; retained as code 1/2/3, not relabeled as sleep stages
WeightKg, WeightPounds, Fat, BMI, IsManualReport	Weight units explicit in field names; Fat sparse; BMI retained as reported; boolean reporting method

Redundancy verification: all three daily subsets agree with `dailyActivity`. Wide/narrow pairs agree on overlapping minute timestamps, but their coverage differs. They must not be appended together or counted as additional users.

```
display(data["redundancy"])
```

	comparison	overlap_rows	missing_keys	mismatched_v alues
0	dailySteps vs dailyActivity	940	0	0
1	dailyCalories vs dailyActivity	940	0	0
2	dailyIntensities vs dailyActivity	940	0	0
3	minuteSteps wide/narrow	1278060	20640	0
4	minuteIntensities wide/narrow	1278060	20640	0
5	minuteCalories wide/narrow	1278060	20640	0

2. Data integrity and quality audit

Takeaway: Missing wear information and uneven modality coverage matter more than nominal row volume.

```
display(data["quality"])
display(data["inventory"][["table_name", "rows", "users", "duplicate_rows", "duplicate_keys", "megabytes"]])
display(data["columns"].query("null_rows > 0 or negative_rows > 0"))
```

	check	count	status	interpretation
0	Participants vs brief	33.0	Review	33 activity users vs 30 stated
1	Exact sleep duplicates	3.0	Fixed	413 raw rows -> 410 unique
2	Minute sleep duplicates	543.0	Fixed	Deduplicated before episode aggregation
3	Zero-step days	77.0	Flagged	May be nonwear or true inactivity
4	Near-zero days (<100)	87.0	Excluded	Proxy threshold; sensitivity table retained
5	Sedentary =1440	79.0	Excluded	24 hours; does not establish actual nonwear
6	Classified minutes >1440	0.0	Review	Impossible full-day duration if present
7	Missing Fat	65.0	Retained	Do not impute sparse body-fat field
8	Multiple sleep records	46.0	Retained	Possible split sleep/naps; no automatic deletion
9	Asleep exceeds in-bed	0.0	Checked	Reject impossible records if present
10	Users with all 31 activity dates	21.0	Review	Unequal observation opportunity
11	Wide format outside target dates	109.0	Review	Wide files are not identical-coverage substitutes
12	Distance and intensity units	NaN	Unresolved	No embedded unit dictionary; raw scores only
13	Demographic fields	0.0	Limitation	No age, sex, height or profession

	table_name	rows	users	duplicate_rows	duplicate_keys	megabytes
0	dailyActivity	940	33	0	0	0.11
1	dailyCalories	940	33	0	0	0.03
2	dailyIntensities	940	33	0	0	0.07
3	dailySteps	940	33	0	0	0.03
4	heartrate_seconds	2483658	14	0	0	89.59
5	hourlyCalories	22099	33	0	0	0.80
6	hourlyIntensities	22099	33	0	0	0.90
7	hourlySteps	22099	33	0	0	0.80
8	minuteCaloriesNarrow	1325580	33	0	0	66.44
9	minuteCaloriesWide	21645	33	0	0	22.99
10	minuteIntensitiesNarrow	1325580	33	0	0	46.36
11	minuteIntensitiesWide	21645	33	0	0	3.31
12	minuteMETsNarrow	1325580	33	0	0	47.69
13	minuteSleep	188521	24	543	543	8.85
14	minuteStepsNarrow	1325580	33	0	0	46.53
15	minuteStepsWide	21645	33	0	0	3.48
16	sleepDay	413	24	3	3	0.02
17	weightLogInfo	67	8	0	0	0.01

	table_name	column_name	storage_type	null_rows	negative_rows
255	weightLogInfo	Fat	REAL	65	0

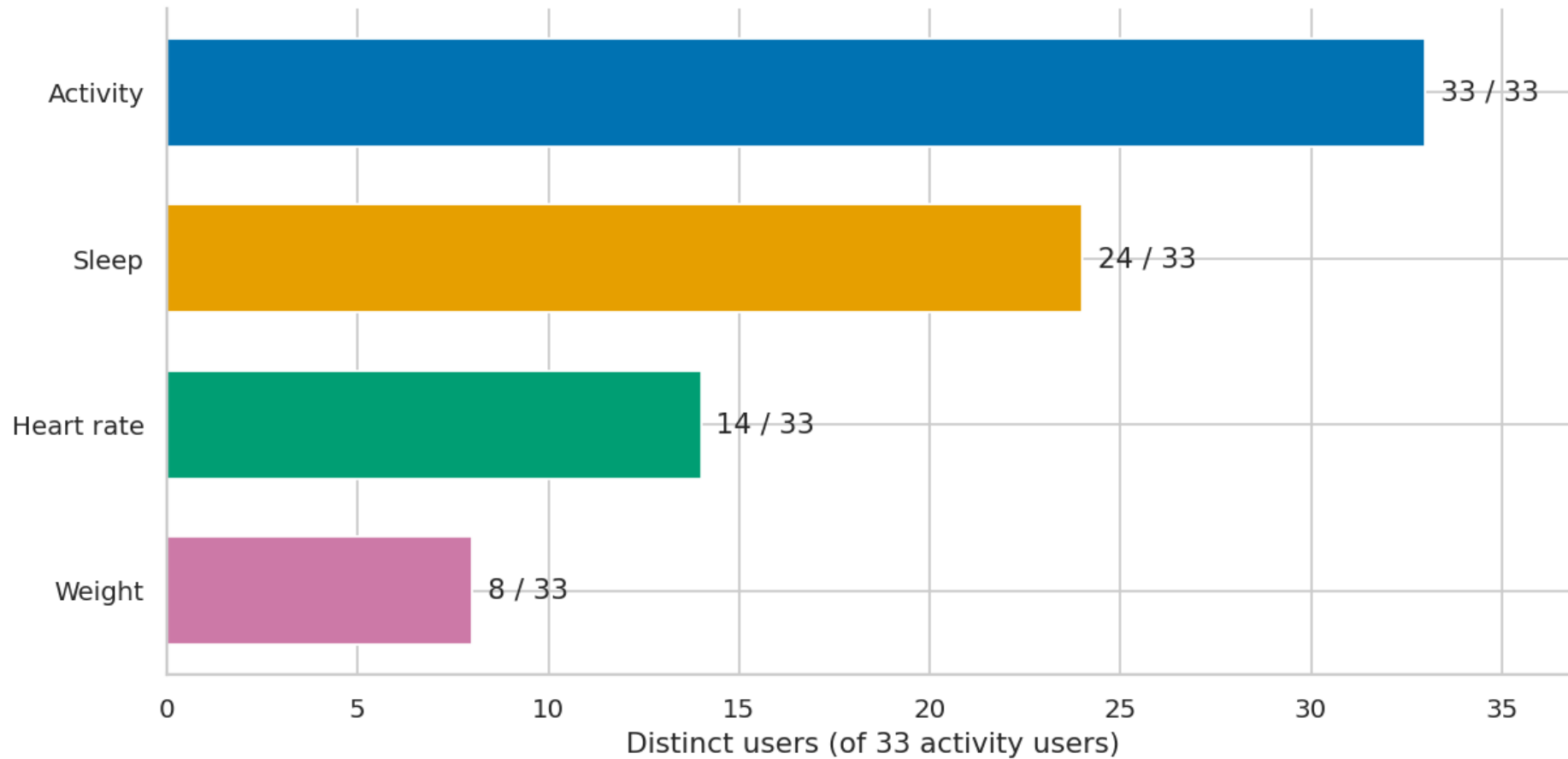
Mismatches against the brief/deck:

- Activity has **33** users rather than 30. Sleep has **24**, heart rate **14**, weight **8**. No demographic restriction to women is possible.
- Daily activity has **940 rows**, daily sleep **413 raw / 410 unique**, weight **67**, hourly steps **22,099**, heart rate **2,483,658** (89.59 MB). The approximate counts supplied are consistent.
- Minute-wide files are only 3.31–22.99 MB; the 45–65 MB description applies mainly to narrow activity exports (minute calories is 66.44 MB in decimal units). `minuteSleep` is 8.85 MB.
- Wide activity extends to **May 13 08:00**, and minute sleep begins **April 11 20:48**. Daily analysis remains April 12–May 12 inclusive, exactly **31 days**. Crossing months does not itself create an inconsistent time frame.

- There are **21** users with all 31 activity dates; users have **4–31** logged dates. Coverage is not a measure of diligence or motivation.
- **77** zero-step days and **79** days with 1,440 sedentary minutes are suspicious, not proof of nonwear. 1,440 minutes equals **24 hours**.
- Weight Fat has **65/67** missing cells, contradicting the deck's blanket “no blanks” claim. Daily sleep has **3** duplicate rows and minute sleep **543**.
- Heart-rate intervals are irregular; fixed 15-second sampling cannot be assumed. A full interval-frequency table is exported in `sql_outputs/hr_sampling_audit.csv`.
- The deck's 10,000-step “CDC requirement” and gender-based calorie-burn conclusions are not supported. The step tiers here are descriptive historical pedometer tiers specified by the task, not a universal health prescription.

```
plot_chart(1, data)
```

01 Weight logging reaches only 8 of 33 activity users

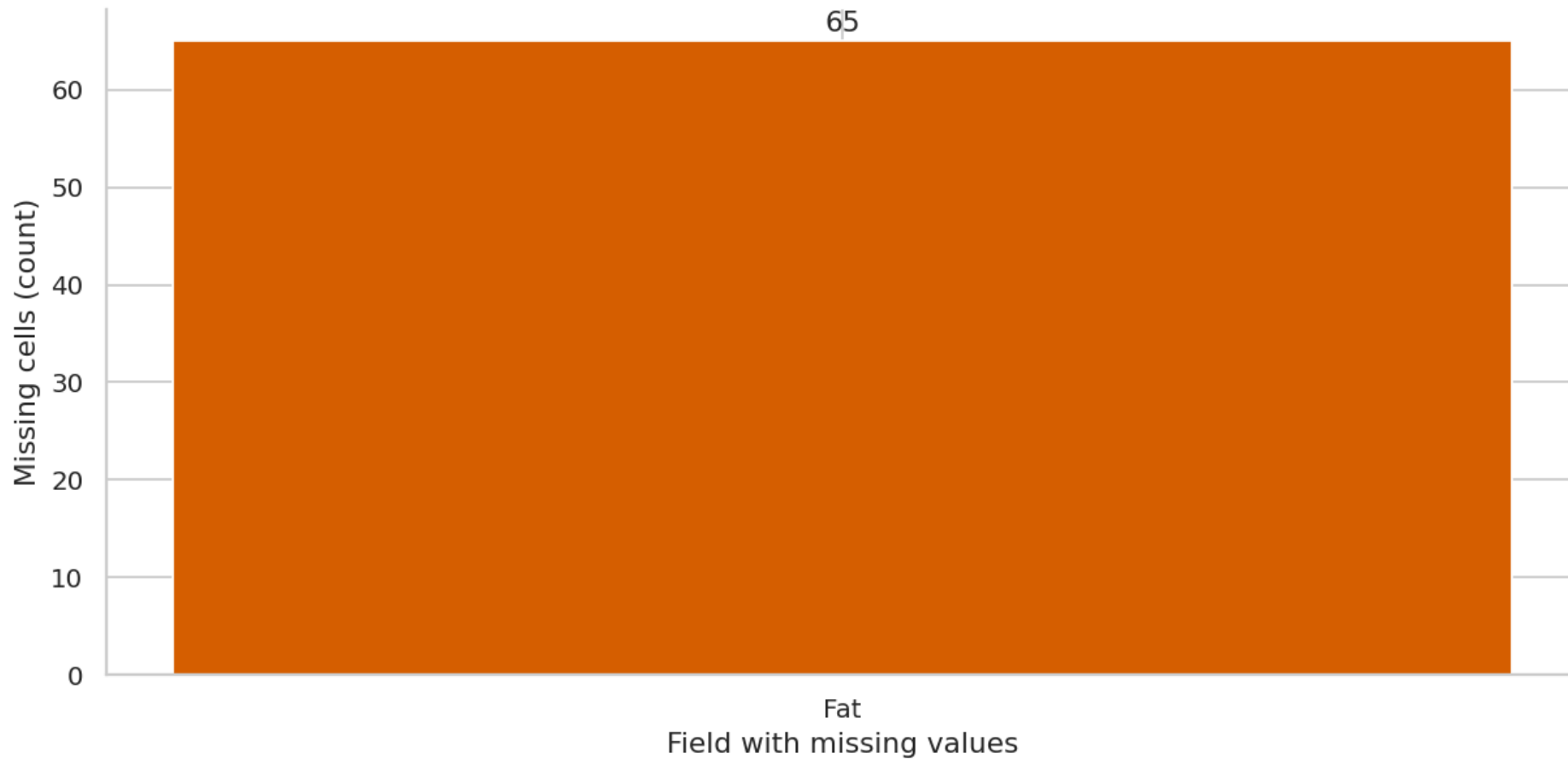


Source: supplied Fitbit / MTurk export, 2016. Descriptive observational analysis; see notebook denominators.

So what? Use activity-led onboarding as the broad entry point. Separate feature adoption counts are not a sequential conversion funnel and do not establish user preference or device capability.

```
plot_chart(2, data)
```

02 65 of 67 weight records omit body-fat percentage



Source: supplied Fitbit / MTurk export, 2016. Descriptive observational analysis; see notebook denominators.

So what? Do not segment or market from body-fat estimates: only 2 of 67 weight logs contain Fat, and no missing value is imputed.

```
display(u[["Id", "first_date", "last_date", "days_logged", "valid_days", "engagement_score"]].sort_values("valid_days"))
display(pd.read_csv(BASE / "sql_outputs/hr_value_audit.csv"))
display(pd.read_csv(BASE / "sql_outputs/hr_sampling_audit.csv").head(12))
```

	id	first_date	last_date	days_logg ed	valid_days	engageme nt_score
13	405719291 2	2016-04-12	2016-04-15	4	3	9.677419
12	402033265 0	2016-04-12	2016-05-12	31	15	48.387097
23	677588895 5	2016-04-12	2016-05-07	26	16	51.612903
4	192797227 9	2016-04-12	2016-05-12	31	17	54.838710
8	234716779 6	2016-04-12	2016-04-29	18	17	54.838710
3	184450507 2	2016-04-12	2016-05-12	31	18	58.064516
28	825324287 9	2016-04-12	2016-04-30	19	18	58.064516
31	879200966 5	2016-04-12	2016-05-10	29	19	61.290323
10	337286816 4	2016-04-12	2016-05-01	20	20	64.516129
21	611766616 0	2016-04-12	2016-05-09	28	23	74.193548
25	700774417 1	2016-04-12	2016-05-07	26	24	77.419355
22	629085500 5	2016-04-12	2016-05-10	29	24	77.419355
30	858381505 9	2016-04-12	2016-05-12	31	25	80.645161
20	557715031 3	2016-04-12	2016-05-11	30	28	90.322581
14	431970357 7	2016-04-12	2016-05-12	31	28	90.322581
26	708636192 6	2016-04-12	2016-05-12	31	29	93.548387
2	164443008 1	2016-04-12	2016-05-11	30	30	96.774194
18	470292168 4	2016-04-12	2016-05-12	31	30	96.774194
11	397733371 4	2016-04-12	2016-05-11	30	30	96.774194
0	150396036 6	2016-04-12	2016-05-12	31	30	96.774194
15	438816184 7	2016-04-12	2016-05-12	31	30	96.774194
1	162458008 1	2016-04-12	2016-05-12	31	31	100.000000
6	202635203 5	2016-04-12	2016-05-12	31	31	100.000000
7	232012700 2	2016-04-12	2016-05-12	31	31	100.000000

	id	first_date	last_date	days_logged	valid_days	engagement_score
9	2873212765	2016-04-12	2016-05-12	31	31	100.000000
5	2022484408	2016-04-12	2016-05-12	31	31	100.000000
19	5553957443	2016-04-12	2016-05-12	31	31	100.000000
16	4445114986	2016-04-12	2016-05-12	31	31	100.000000
17	4558609924	2016-04-12	2016-05-12	31	31	100.000000
24	6962181067	2016-04-12	2016-05-12	31	31	100.000000
27	8053475328	2016-04-12	2016-05-12	31	31	100.000000
29	8378563200	2016-04-12	2016-05-12	31	31	100.000000
32	8877689391	2016-04-12	2016-05-12	31	31	100.000000

	readings	min_bp_m	max_bp_m	out_of_bounds
0	2483658	36.0	203.0	0

	seconds	intervals
0	5.0	868956
1	10.0	622040
2	15.0	557885
3	3.0	245262
4	1.0	120929
5	2.0	65203
6	6.0	697
7	7.0	277
8	45.0	213
9	4.0	196
10	8.0	129
11	35.0	85

3. Cleaning and feature engineering

Takeaway: The transparent quality proxy retains 846 of 940 daily records; the raw series remains available.

```
display(data["cleaning"])
display(data["sensitivity"])
```

		issue	action	rows_before	rows_after	columns_before	columns_after	justification
	0	Exact activity duplicates	Deduplicate activity	940	940	15	15	Retain one identical record
	1	Probable nonwear / negligible motion	Apply wear-quality proxy	940	846	17	17	Steps >=100 and sedentary <1440; excludes true low movement too; sensitivity reported
	2	Analytical features	Engineer daily features	846	846	17	26	Active share denominator is classified minutes; calories include basal expenditure
	3	Exact sleep duplicates	Deduplicate sleep	413	410	5	5	Remove identical daily records
	4	Impossible sleep values	Validate sleep	410	410	5	5	Positive duration, asleep <= in-bed <=1440
	5	Weight validity and missing Fat	Validate weight; retain missing Fat	67	67	8	8	No imputation; analyse logging, not weight change or health
	6	Hourly coverage	Restrict hours to retained activity dates	22099	20045	3	11	No imputation for absent hours; averages use observed user-hours
	7	Duplicate minute sleep rows	Aggregate unique sleep minutes to episodes	188521	459	4	8	Exact deduplication in SQL; raw code labels not interpreted as clinical stages
	8	Repeated user-days	Aggregate user features and attach personas	846	33	27	17	One observation per user for clustering and inference
	9	Calendar alignment	Join sleep to exact adjacent activity dates	410	410	11	6	Left joins preserve sleep days; unmatched activity remains missing
	10	Incomplete user-weeks	Aggregate complete study windows	846	125	27	7	Exclude final 3-day fragment; flags require seven retained days
	11	Sparse overlapping sensors	Join hourly HR, intensity and raw METs	6013	5990	6	9	Inner one-to-one joins; retain only overlapping hours on proxy-valid activity dates

	minimu m_steps	days	mean_st eps
0	1	856	8328.69 3925
1	100	846	8426.83 2151
2	500	835	8534.58 3234
3	1000	824	8638.45 3883

Cleaning contract: retain exact-unique daily activity rows with **TotalSteps** ≥ 100 and **SedentaryMinutes** $< 1,440$. This operational proxy excludes plausible nonwear and near-zero motion; it also removes some genuinely inactive days. Never label it confirmed wear time. No extra calorie or distance exclusion is applied. Preserve all raw rows with the flag in

`flagged_daily_activity`.

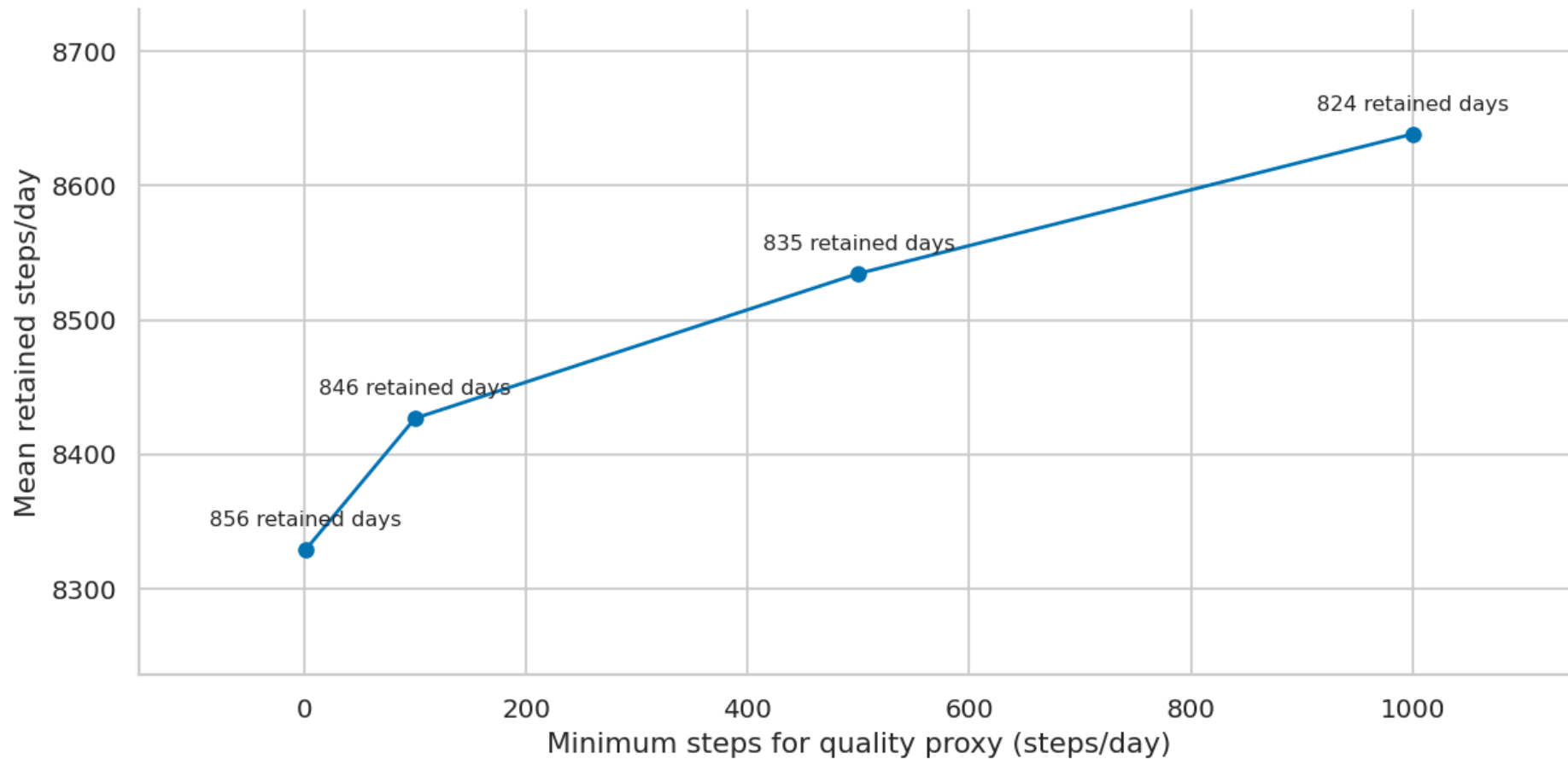
Exact-duplicate sleep rows are removed. Retain positive sleep with asleep $\leq$ in-bed $\leq 1,440$. Sleep results use all valid sleep records, not only activity-filtered dates. Retain positive WeightKg and BMI, leaving Fat missing. Multiple weight logs remain separate; aggregate to user-day before daily joins. Unique minute-sleep records are aggregated to episodes. HR summaries use 30–220 bpm as a broad plausibility screen; intervals over 60 seconds and final readings with no following observation contribute no duration.

Features: weekday/weekend, study day/week, four hour buckets; five requested step tiers; active minutes = light + fairly + very active; active share = active / (active + sedentary classified minutes); sleep efficiency = asleep / in-bed; calories per step = total kcal / steps (not energy cost per step); days logged; engagement = retained days / 31 $\times 100$. Full-window engagement intentionally penalizes missing enrollment days because enrollment dates are unavailable.

WHO-inspired flag: sum fairly-active minutes + 2 $\times$ very-active minutes across each of four complete seven-day study windows starting April 12. Only a user-week with all seven retained days is eligible; threshold ≥ 150 moderate-equivalent minutes. A separate moderate-only ≥ 150 -minute flag is also exported. The final three days are excluded from weekly flags. Fitbit categories are imperfect intensity proxies; strength training and other guideline components cannot be assessed. This is **not a verified health-guideline compliance measure**. [WHO physical activity guidance](#).

```
plot_chart(3, data)
```

03 Higher wear-proxy thresholds selectively raise the activity baseline



Source: supplied Fitbit / MTurk export, 2016. Descriptive observational analysis; see notebook denominators.

So what? The main rule removes 94 days (10.0%). Raw mean steps are 7,638; the retained mean is 8,427. This increase is selection, not an intervention effect.

4. Univariate EDA

Takeaway: Typical activity and recording consistency are different dimensions of device use.

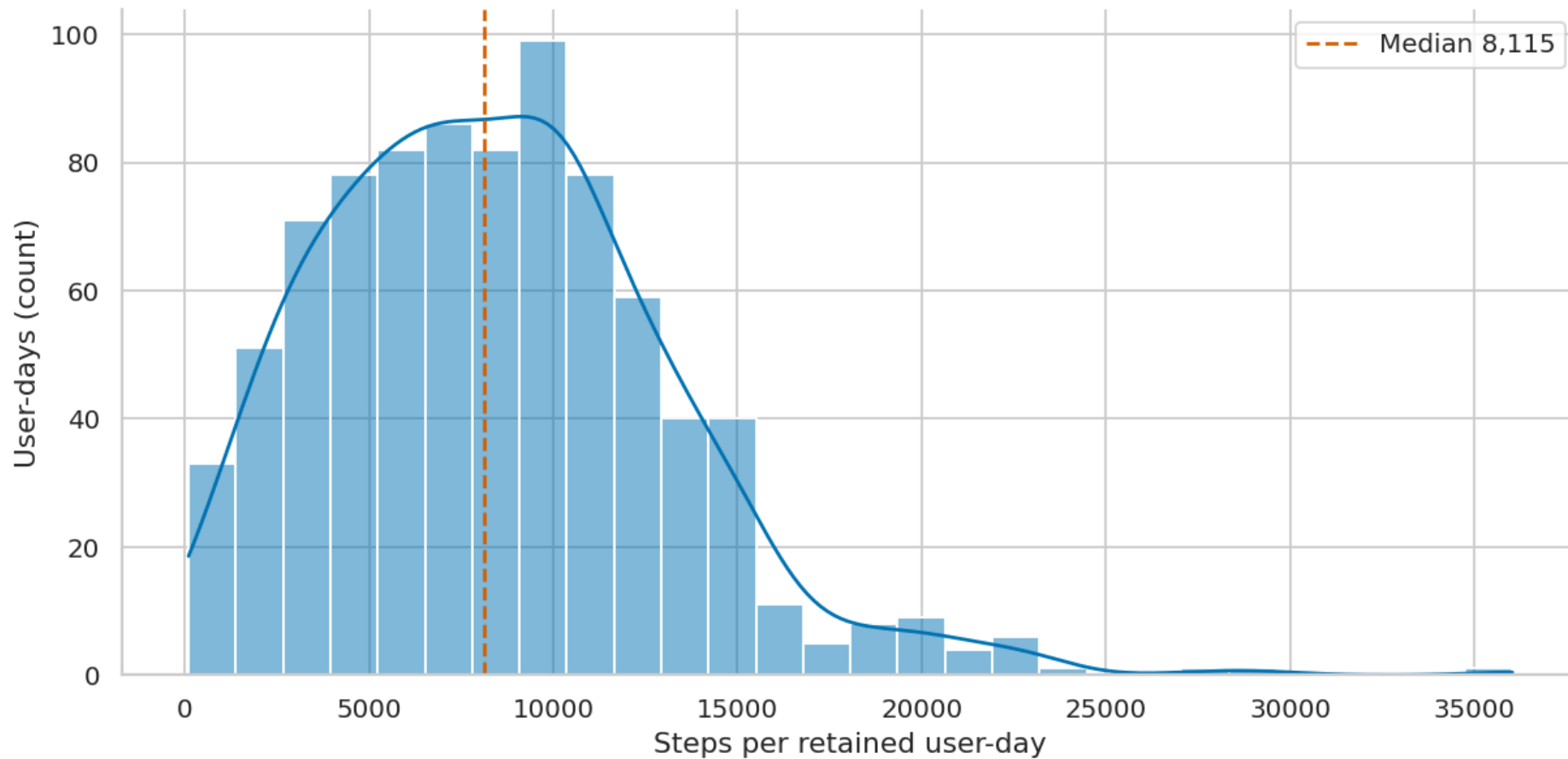
```
display(d[["TotalSteps", "Calories", "active_minutes", "SedentaryMinutes", "active_share"]].describe().round(2))
display(s[["TotalMinutesAsleep", "sleep_efficiency", "in_bed_gap"]].describe().round(2))
```

	TotalSteps	Calories	active_min utes	Sedentary Minutes	active_shar e
count	846.00	846.00	846.00	846.00	846.00
mean	8426.83	2370.38	252.76	949.53	0.22
std	4695.87	697.26	100.55	271.81	0.10
min	108.00	52.00	3.00	13.00	0.00
25%	4981.75	1861.50	190.00	720.25	0.16
50%	8115.00	2222.50	259.00	1017.50	0.23
75%	11128.00	2835.00	324.00	1182.00	0.28
max	36019.00	4900.00	552.00	1437.00	0.61

	TotalMinutes Asleep	sleep_efficie ncy	in_bed_gap
count	410.00	410.00	410.00
mean	419.17	0.92	39.31
std	118.64	0.09	46.71
min	58.00	0.50	0.00
25%	361.00	0.91	17.00
50%	432.50	0.94	25.50
75%	490.00	0.96	40.00
max	796.00	1.00	371.00

```
plot_chart(4, data)
```

04 45.4% of retained days stay below 7,500 steps

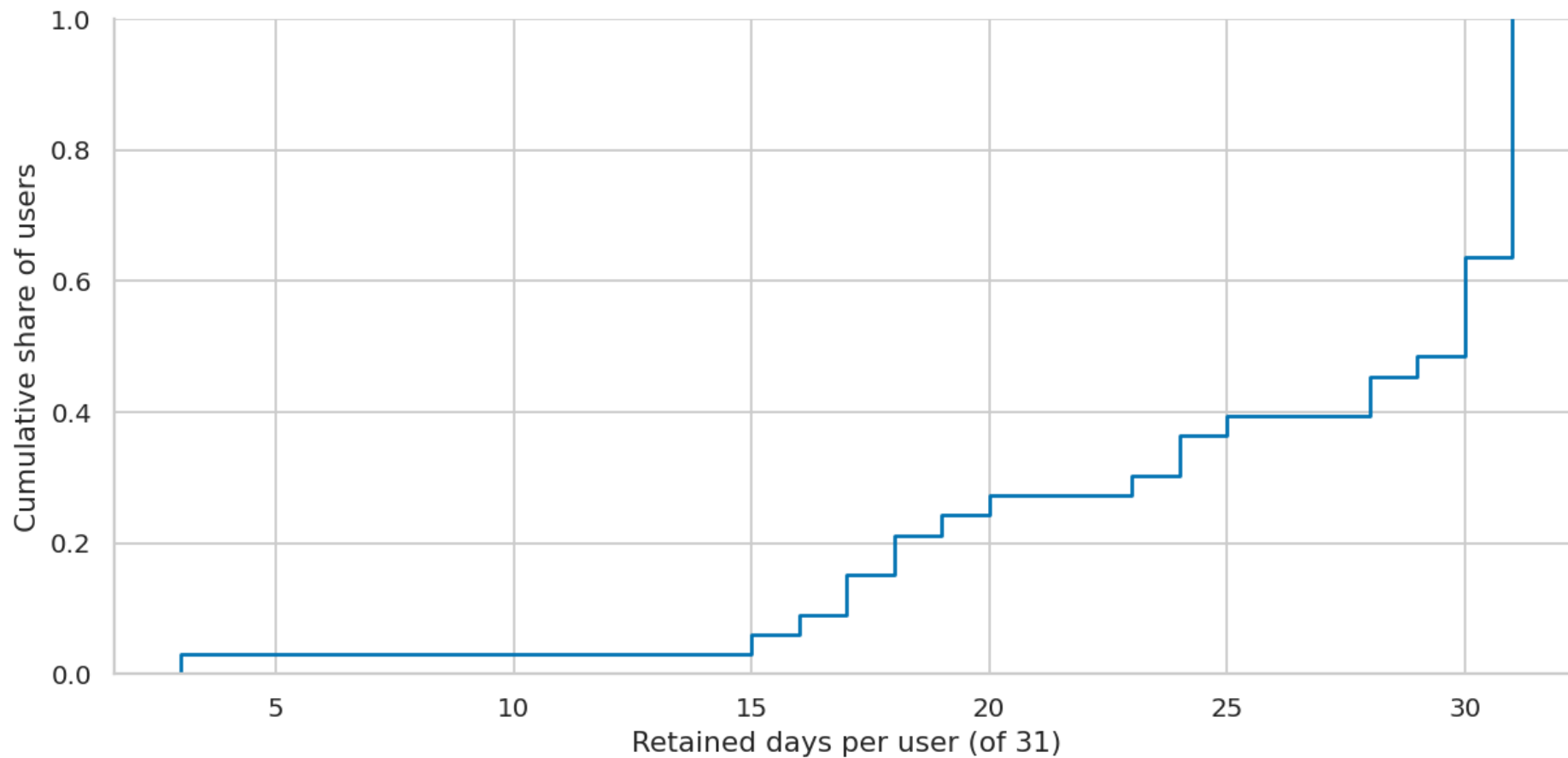


Source: supplied Fitbit / MTurk export, 2016. Descriptive observational analysis; see notebook denominators.

So what? 45.4% of retained days are below 7,500 steps. Set an individual baseline before offering a stretch goal; the tiers do not diagnose health.

```
plot_chart(5, data)
```

05 Users contribute 3 to 31 retained activity days

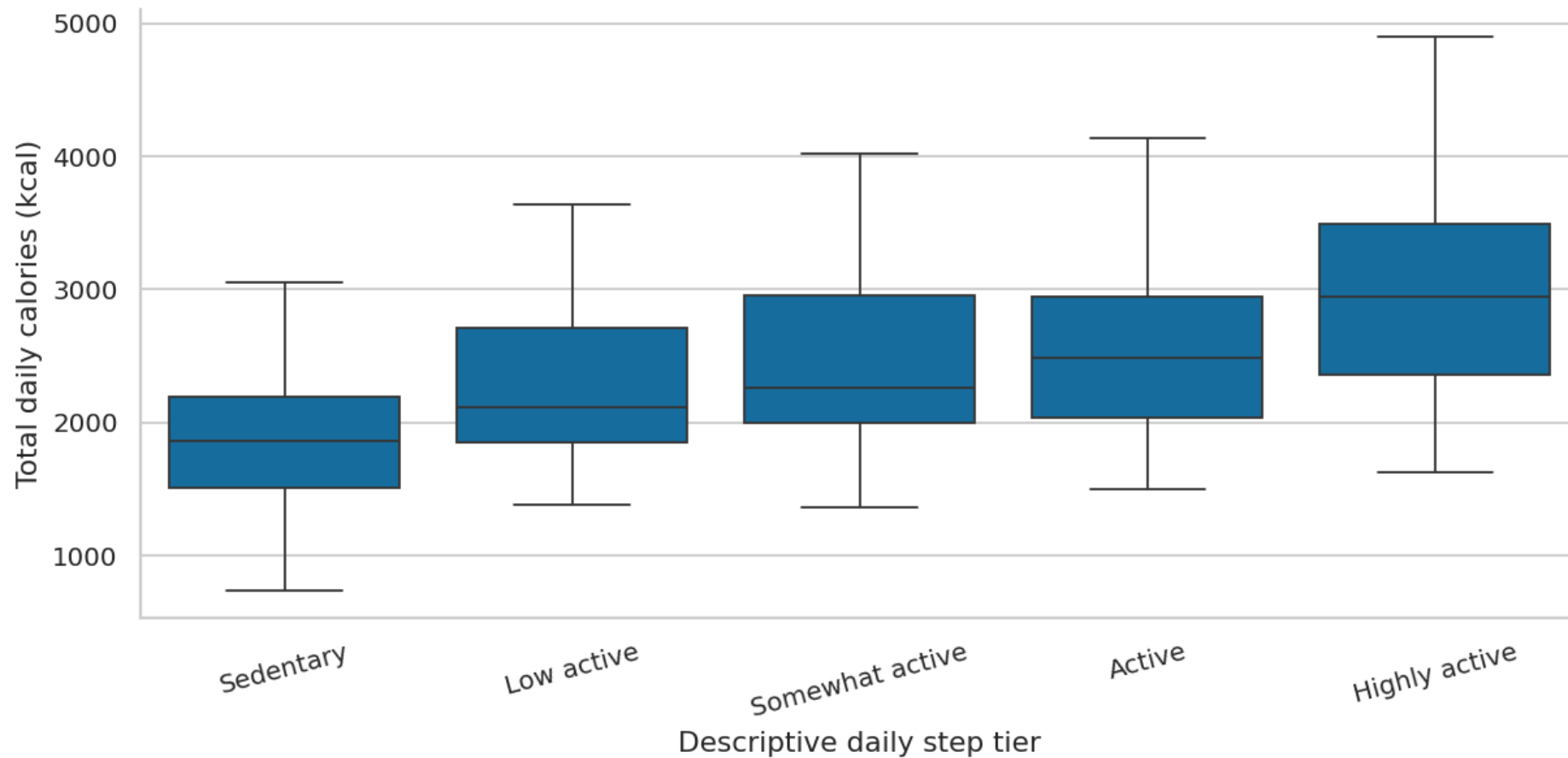


Source: supplied Fitbit / MTurk export, 2016. Descriptive observational analysis; see notebook denominators.

So what? Every user has at least one retained day, but a common 31-day denominator reveals uneven recording opportunity. Logged rows alone overstate usable activity coverage.

```
plot_chart(6, data)
```

06 Total calorie distributions overlap across step tiers



Source: supplied Fitbit / MTurk export, 2016. Descriptive observational analysis; see notebook denominators.

So what? Calories are daily total expenditure estimates and vary between people. These records cannot establish calorie intake, an energy deficit or a gender-specific target.

```
plot_chart(7, data)
```

07 Weekday and weekend activity have substantial within-group variation



Source: supplied Fitbit / MTurk export, 2016. Descriptive observational analysis; see notebook denominators.

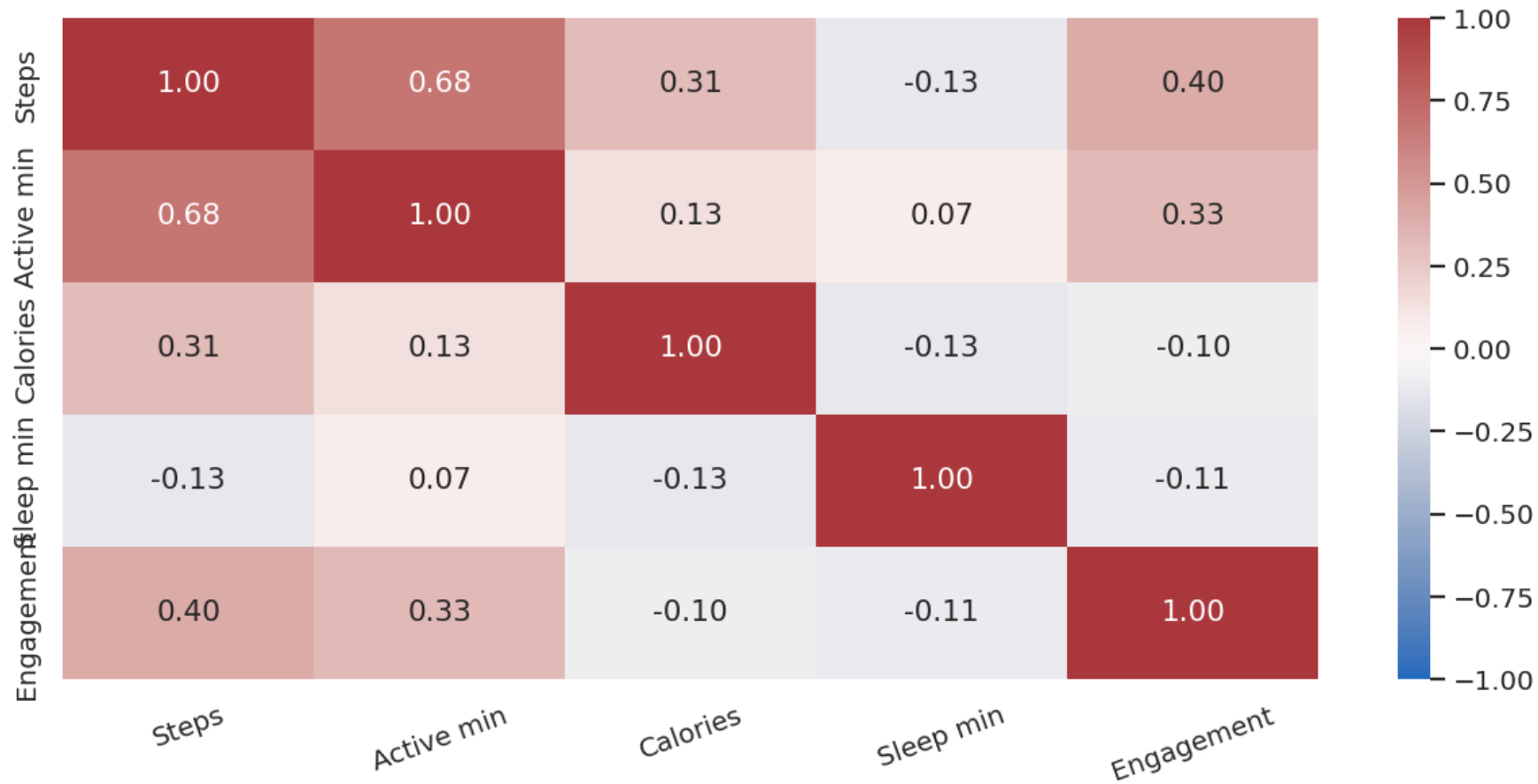
So what? Wide overlapping daily distributions argue against one universal weekend reminder. Repeated days are descriptive here; the formal weekend comparison pairs user means.

5. Bivariate and multivariate EDA

Takeaway: Daily associations should not be mistaken for independent evidence from hundreds of people.

```
plot_chart(9, data)
```

09 User-level associations separate typical behaviour from repeated daily records

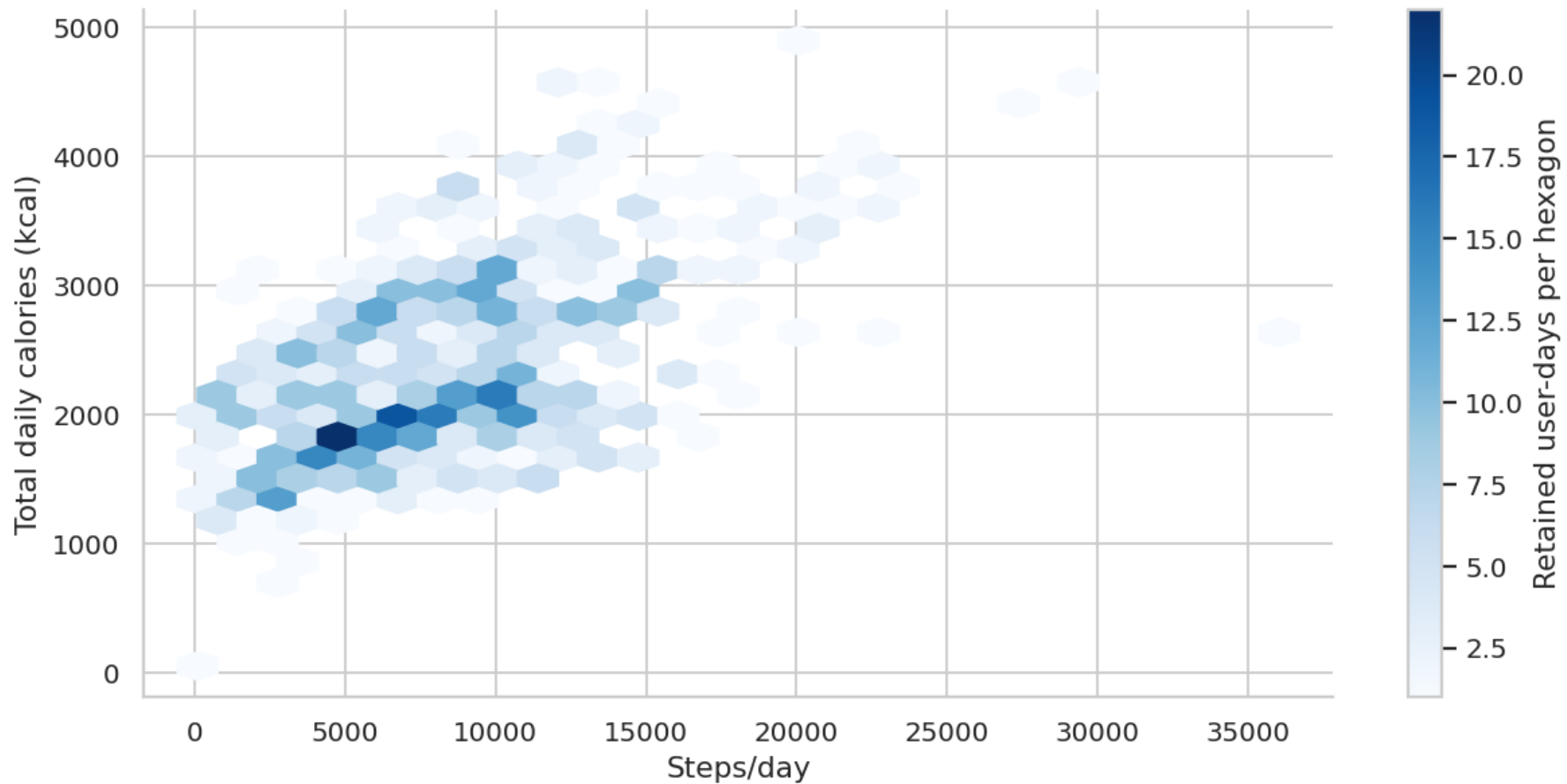


Source: supplied Fitbit / MTurk export, 2016. Descriptive observational analysis; see notebook denominators.

So what? Pairwise correlations use one mean per user. Sleep pairs involve only the 24 sleep loggers; differences in observation days still limit comparability.

```
plot_chart(10, data)
```

10 Similar step totals correspond to different daily calorie estimates

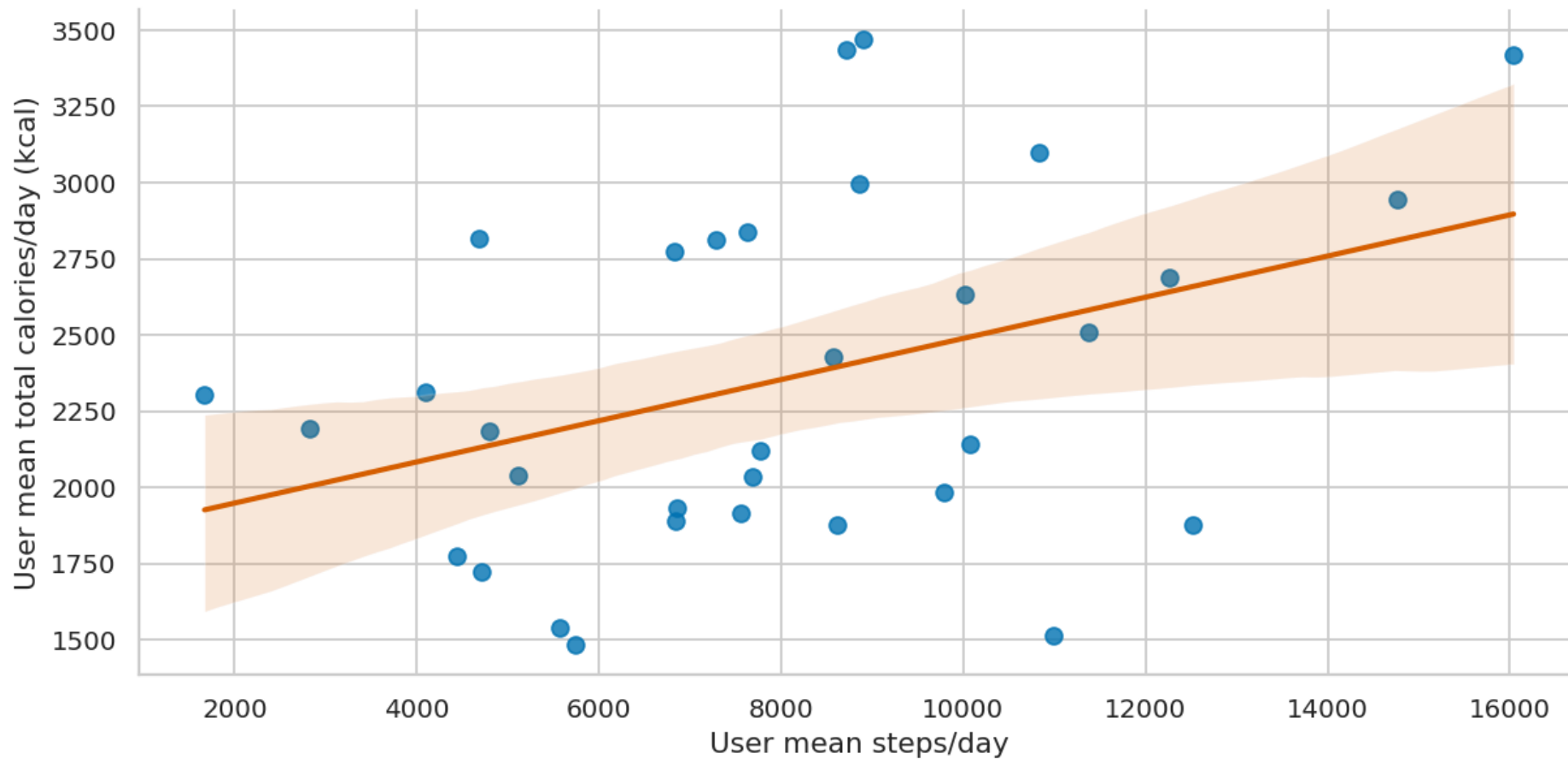


Source: supplied Fitbit / MTurk export, 2016. Descriptive observational analysis; see notebook denominators.

So what? Steps alone cannot explain total calorie differences. Basal expenditure, unmeasured body characteristics and non-step activities remain potential confounders.

```
plot_chart(11, data)
```

11 Between-user calorie association is descriptive, with a 95% bootstrap band



Source: supplied Fitbit / MTurk export, 2016. Descriptive observational analysis; see notebook denominators.

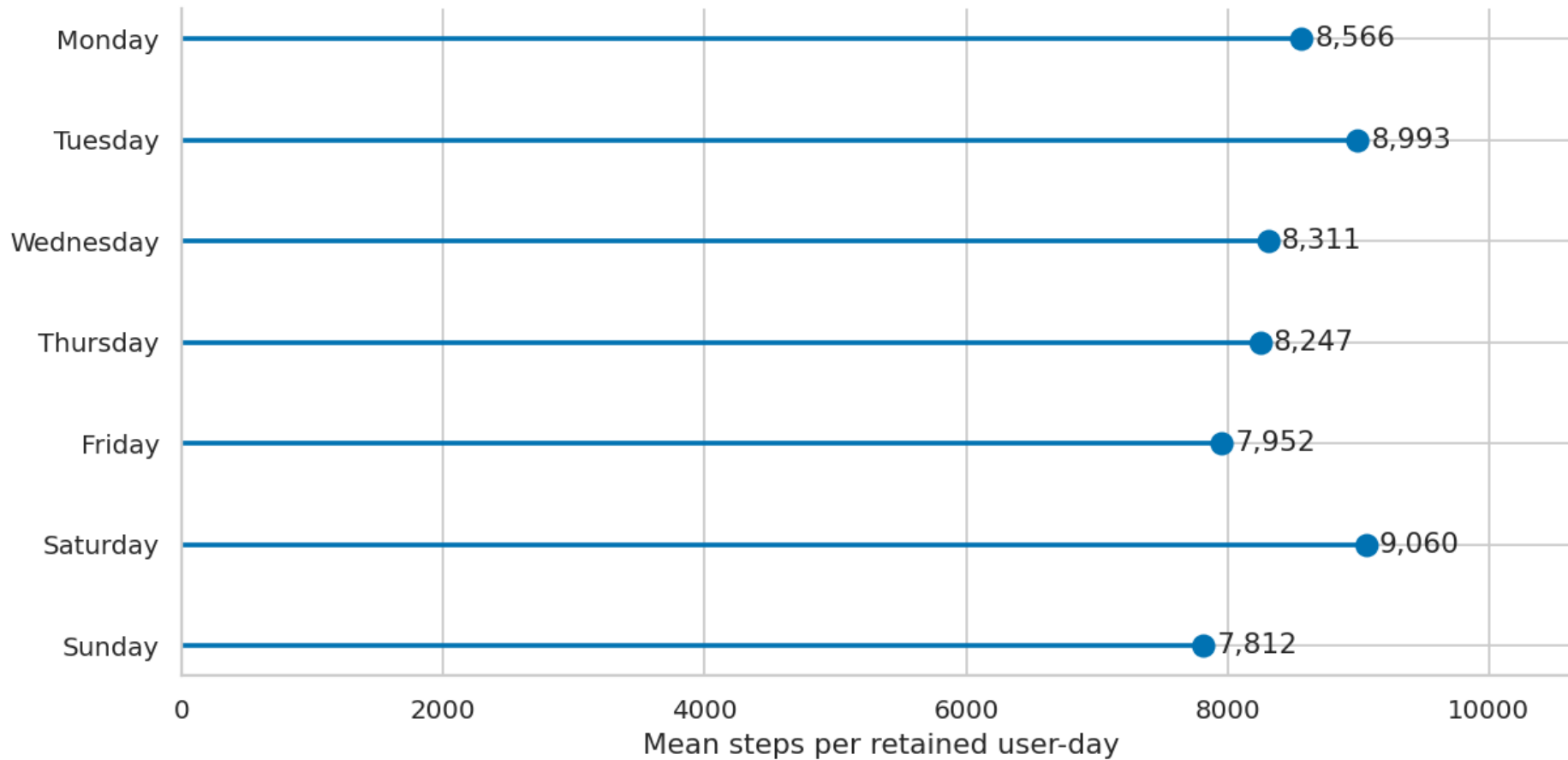
So what? The shaded interval resamples user summaries. With 33 users, a visible fitted trend is insufficient to support a causal or personalized calorie claim.

6. Time and behaviour patterns

Takeaway: Use observed timing to design experiments, while distinguishing calendar coverage from retention.

```
plot_chart(12, data)
```

12 Saturday has the highest pooled mean at 9,060 steps

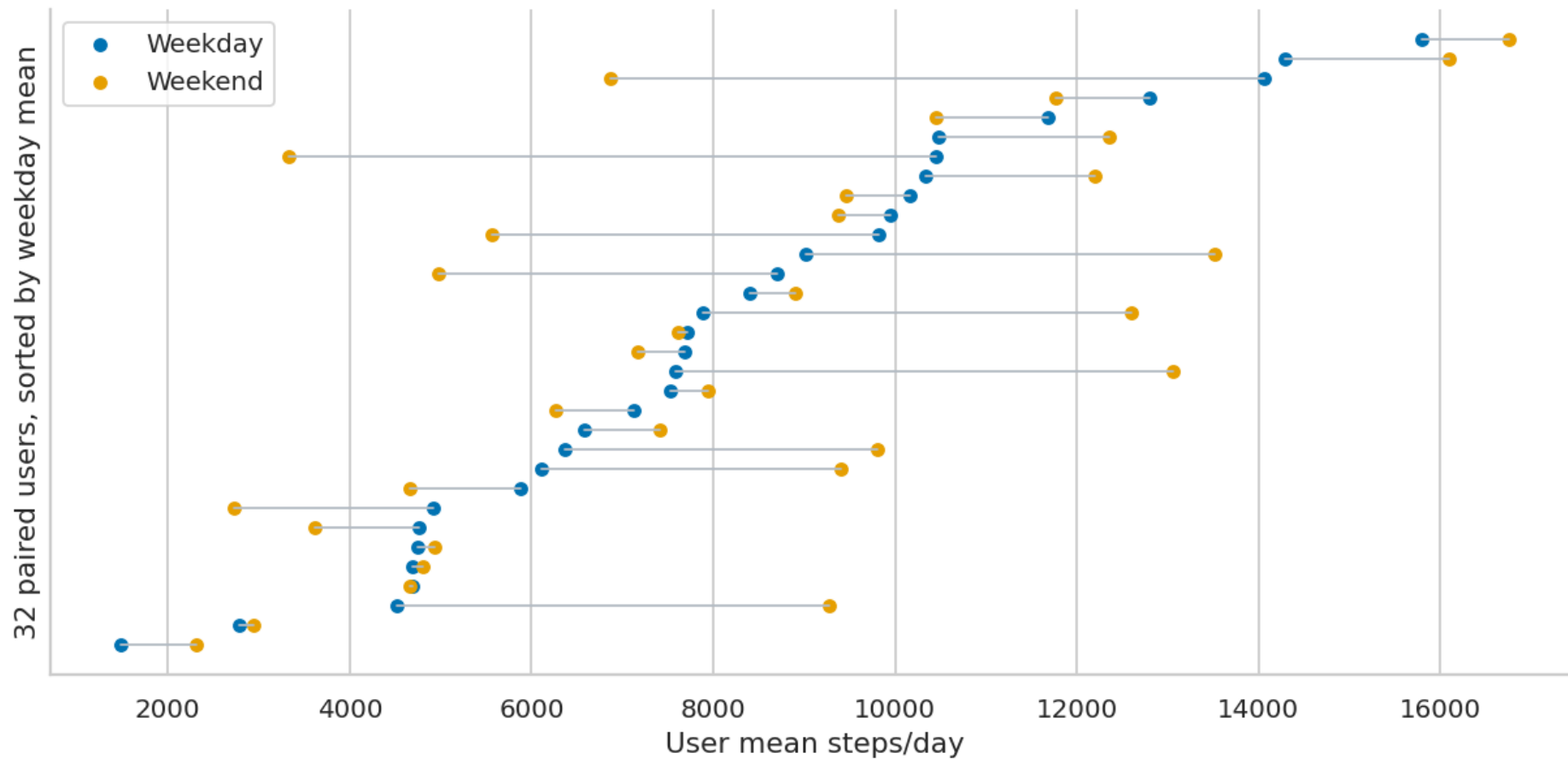


Source: supplied Fitbit / MTurk export, 2016. Descriptive observational analysis; see notebook denominators.

So what? The weekday ranking pools retained days. Different users and different counts contribute on each weekday, so it is not a within-person causal comparison.

```
plot_chart(13, data)
```

13 The average paired weekend gap is +120 steps

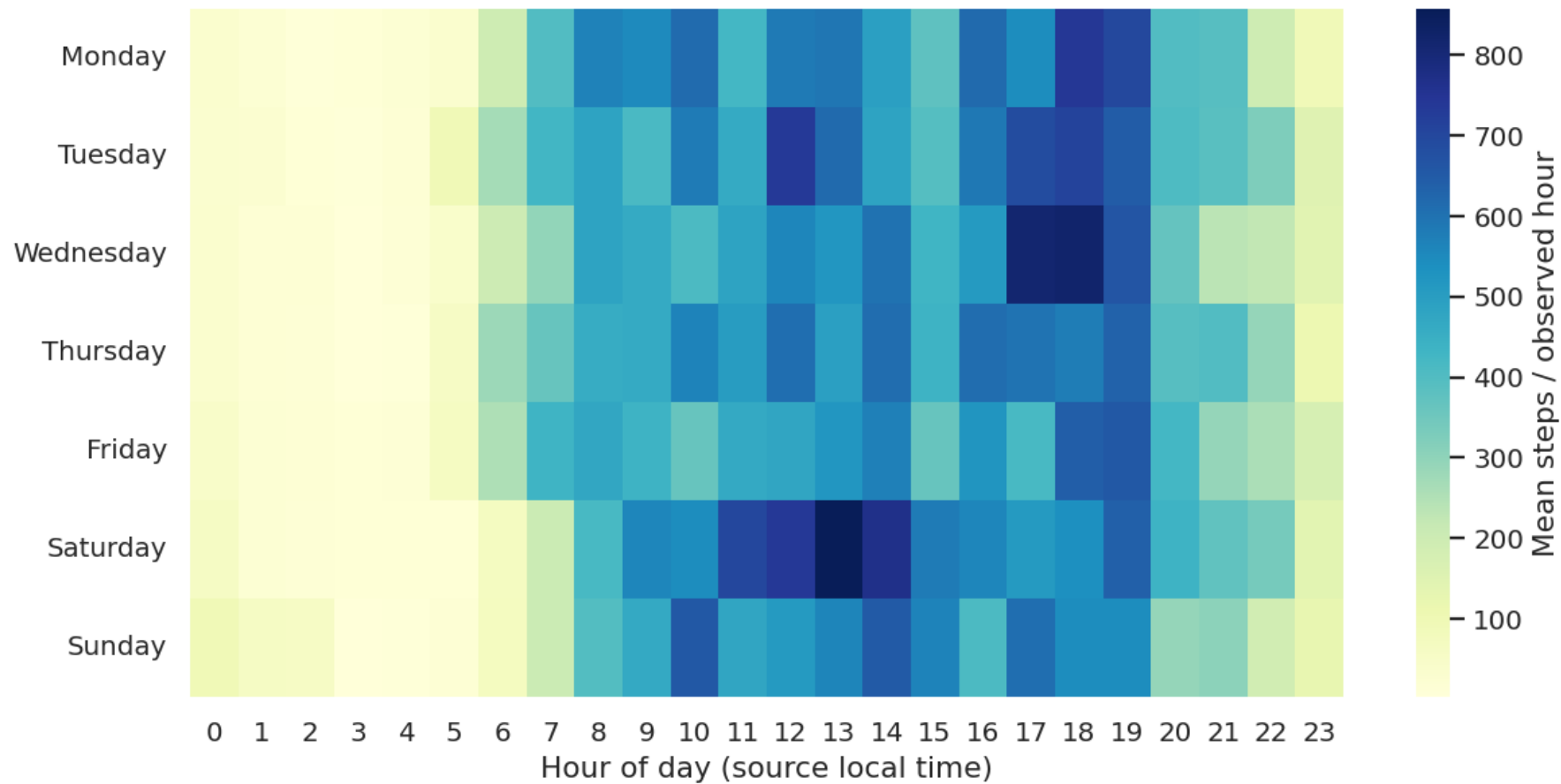


Source: supplied Fitbit / MTurk export, 2016. Descriptive observational analysis; see notebook denominators.

So what? Among 32 users observed in both groups, the mean weekend-minus-weekday gap is +120 steps (95% user-bootstrap CI -959 to +1143). The interval includes zero.

```
plot_chart(14, data)
```

14 Observed activity peaks at 18:00 overall

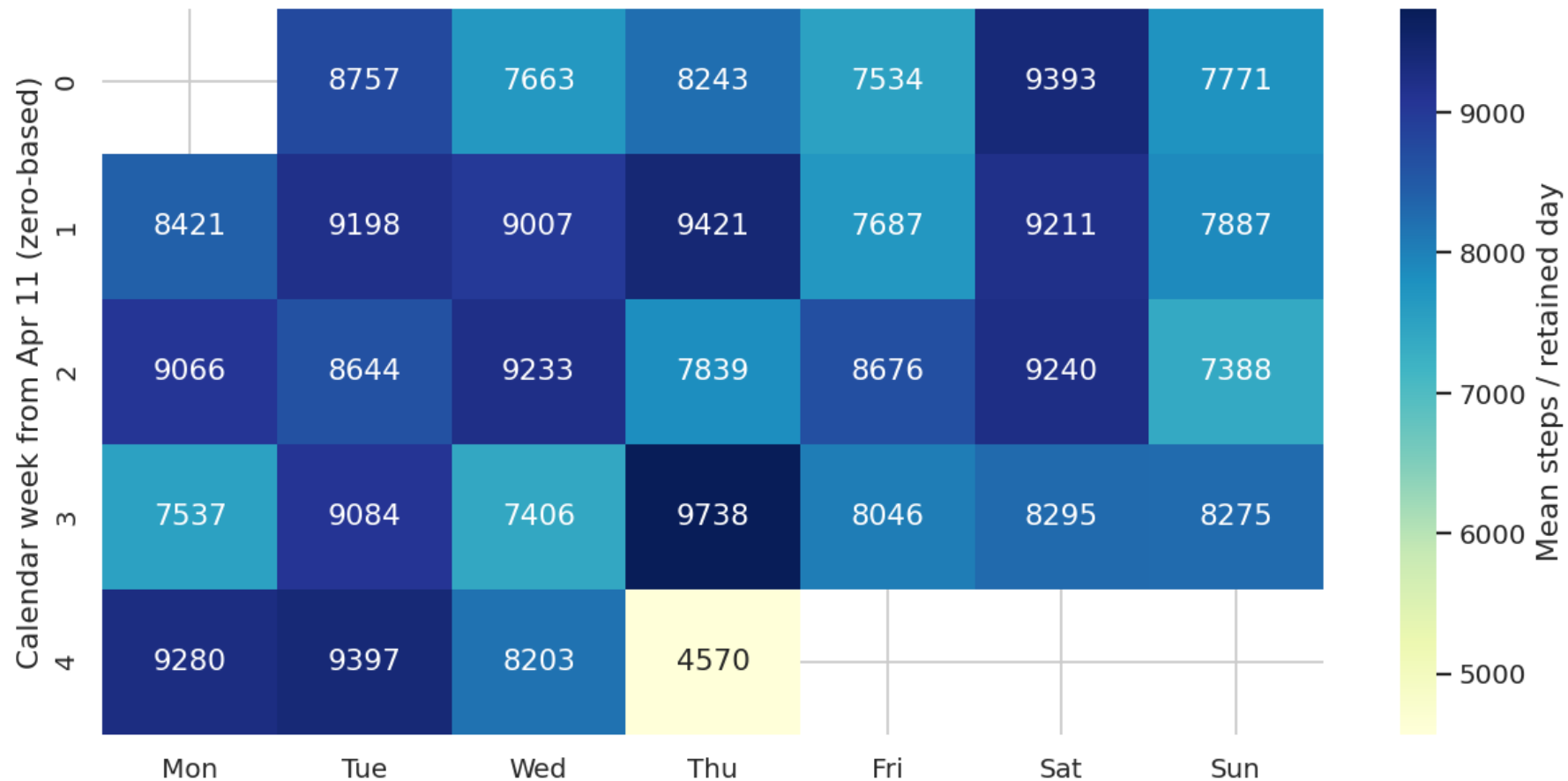


Source: supplied Fitbit / MTurk export, 2016. Descriptive observational analysis; see notebook denominators.

So what? The overall peak is 18:00, averaging 660 steps per observed retained-day hour. Test reminder timing around individual patterns; recorded peak movement is not proven optimal notification timing.

```
plot_chart(15, data)
```

15 Calendar coverage makes the partial final week visible

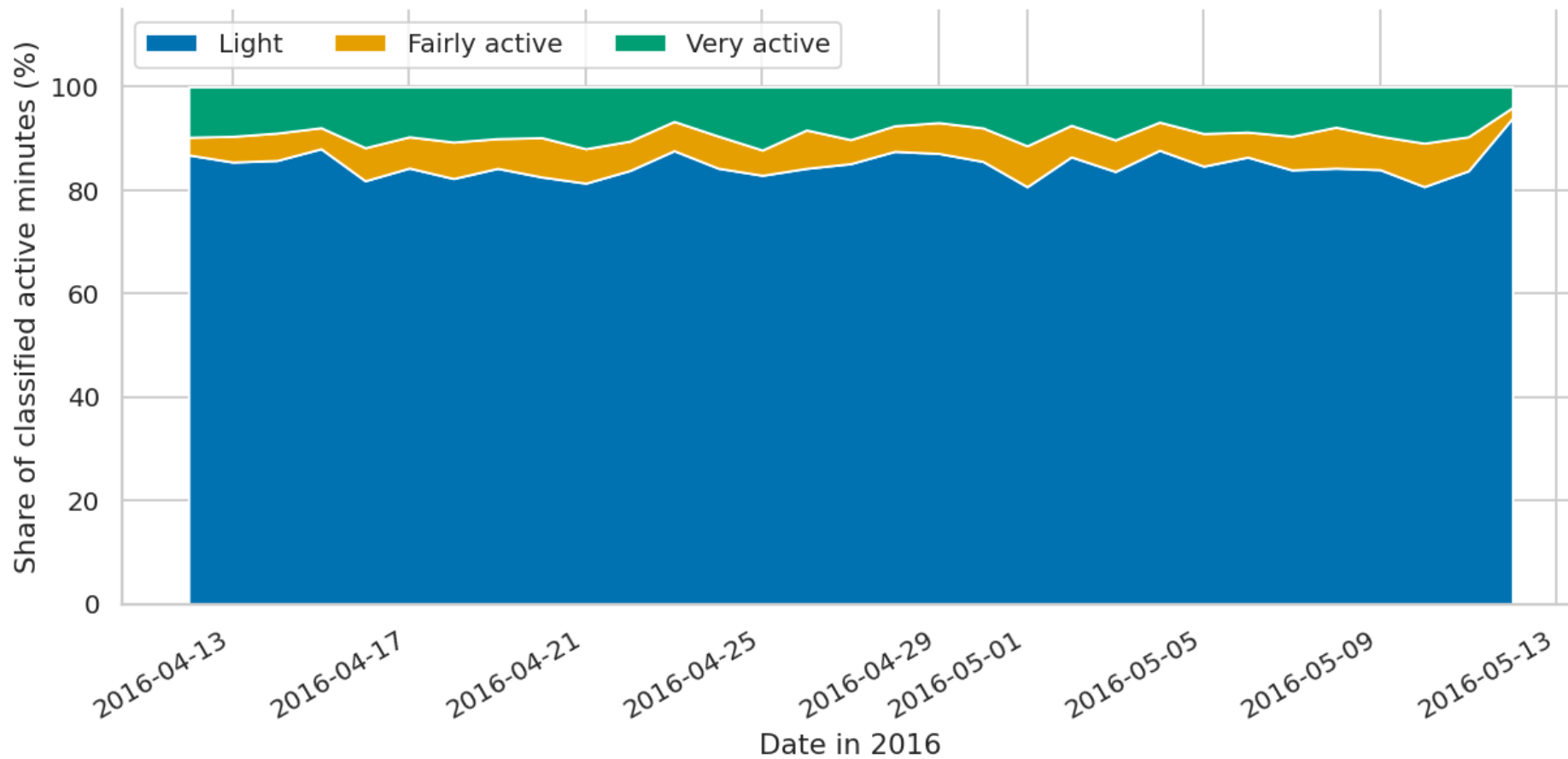


Source: supplied Fitbit / MTurk export, 2016. Descriptive observational analysis; see notebook denominators.

So what? Blank cells fall outside the study window. The partially observed final day and changing user mix can depress daily means, so do not call the pattern behavioural decline.

```
plot_chart(16, data)
```

16 Light activity contributes 84.7% of active minutes

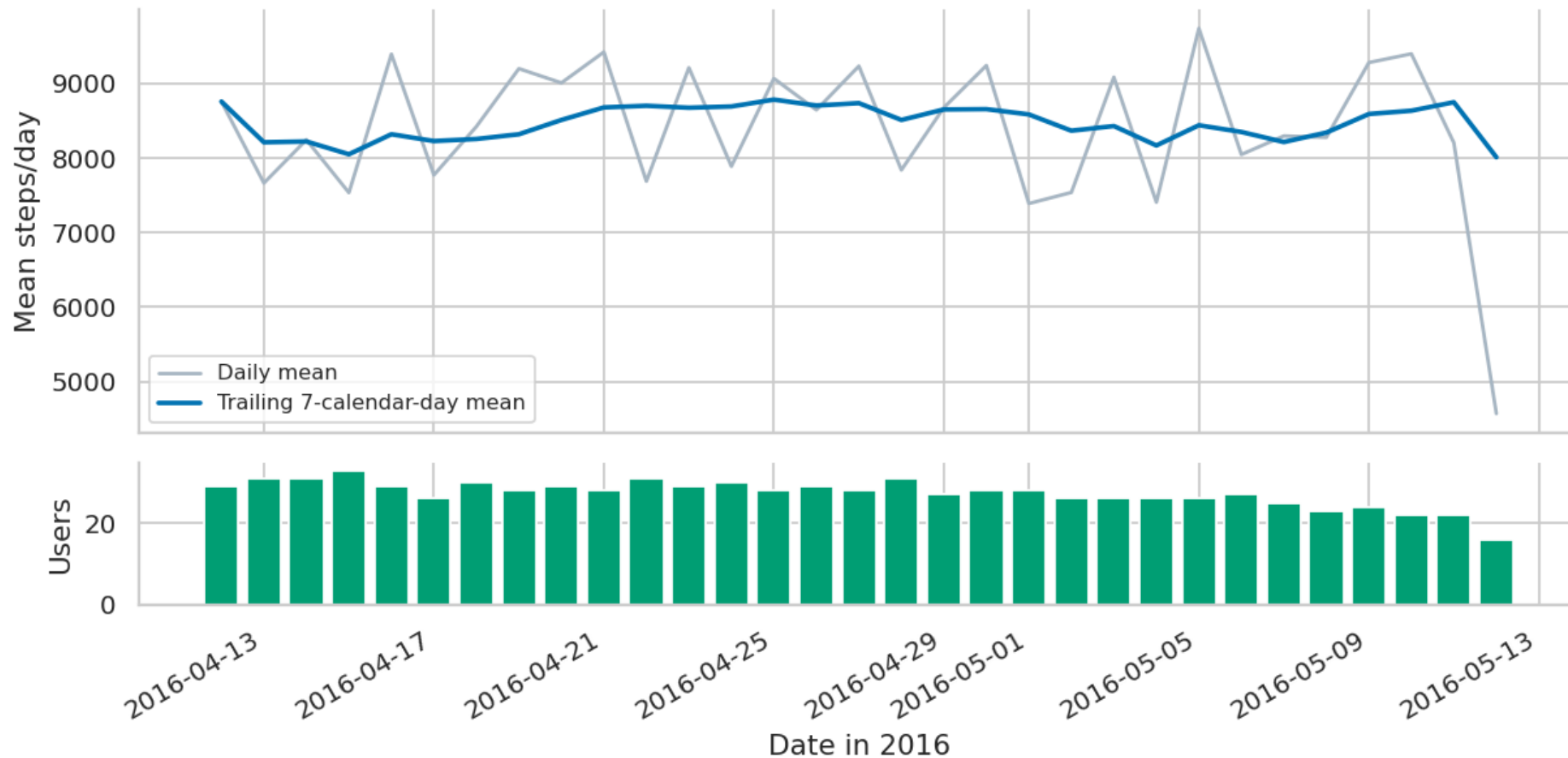


Source: supplied Fitbit / MTurk export, 2016. Descriptive observational analysis; see notebook denominators.

So what? Light minutes account for 84.7% of classified active time. Short movement prompts are a relevant test; light activity is not substituted for moderate intensity in the weekly flag.

```
plot_chart(17, data)
```

17 Changing participant coverage complicates apparent month-long trends



Source: supplied Fitbit / MTurk export, 2016. Descriptive observational analysis; see notebook denominators.

So what? The lower panel shows the changing denominator. A trailing mean smooths daily variation but does not correct attrition or partial-day measurement.

```
display(pd.read_csv(BASE / "sql_outputs/weekly_retention.csv"))
display(pd.read_csv(BASE / "sql_outputs/who_guideline.csv"))
```

	study_week	observable_users	valid_days
0	1	33	209
1	2	31	203
2	3	32	197
3	4	29	177
4	5	22	60

	study_week	observed_user_weeks	eligible_weeks	meeting_proxy	moderate_only_meeting
0	1	33	23	15	4
1	2	31	25	19	8
2	3	32	22	18	7
3	4	29	21	16	4

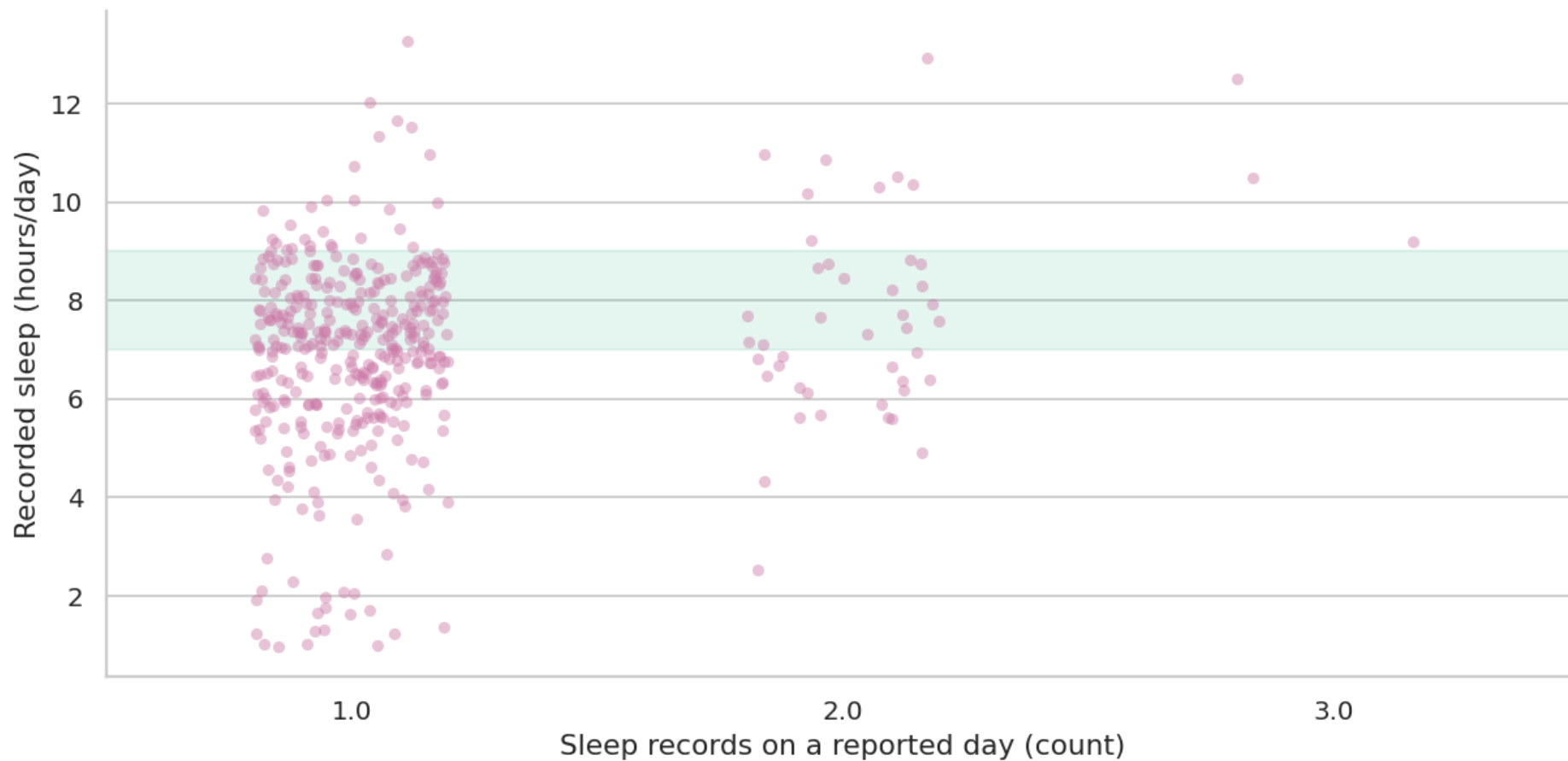
Weekly benchmark: 68 of 91 fully retained user-weeks (74.7%) reach 150 moderate-equivalent minutes; only 23 reach 150 fairly-active minutes alone. These are repeated **user-weeks**, not independent people. Missing/filtered days produce unknown flags rather than automatic failure. Weekly observable-user counts are a retention-style view of this fixed study cohort, not product D7/D30 retention.

7. Sleep analysis

Takeaway: Recorded sleep is close to seven hours on average, but coverage and daily variation matter.

```
plot_chart(8, data)
```

08 46 sleep days combine more than one record

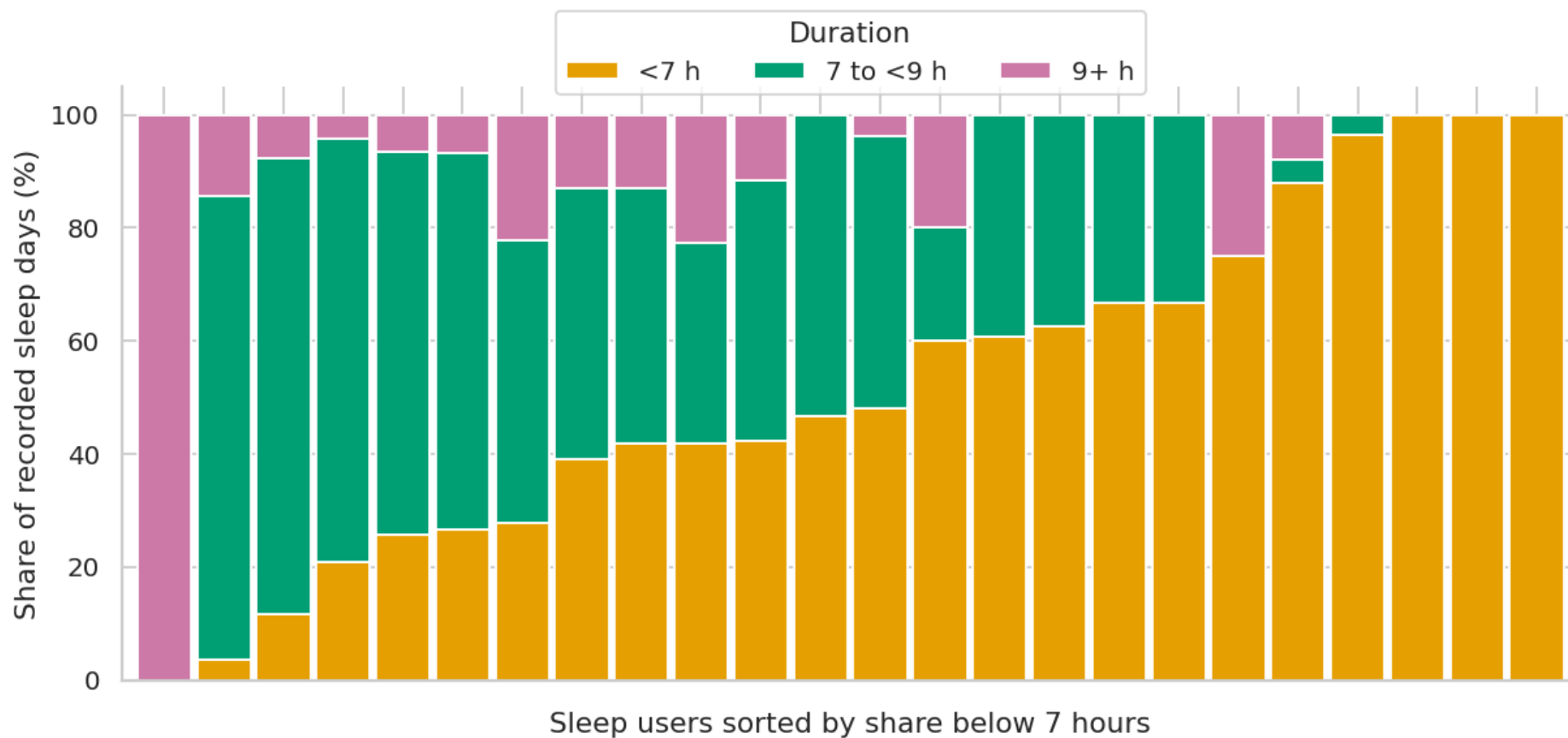


Source: supplied Fitbit / MTurk export, 2016. Descriptive observational analysis; see notebook denominators.

So what? 46 of 410 unique sleep days contain multiple records. Their total may include naps or split sleep, so daily totals must not be presented as uninterrupted nightly sleep.

```
plot_chart(18, data)
```

18 44.1% of recorded sleep days fall below 7 hours



Source: supplied Fitbit / MTurk export, 2016. Descriptive observational analysis; see notebook denominators.

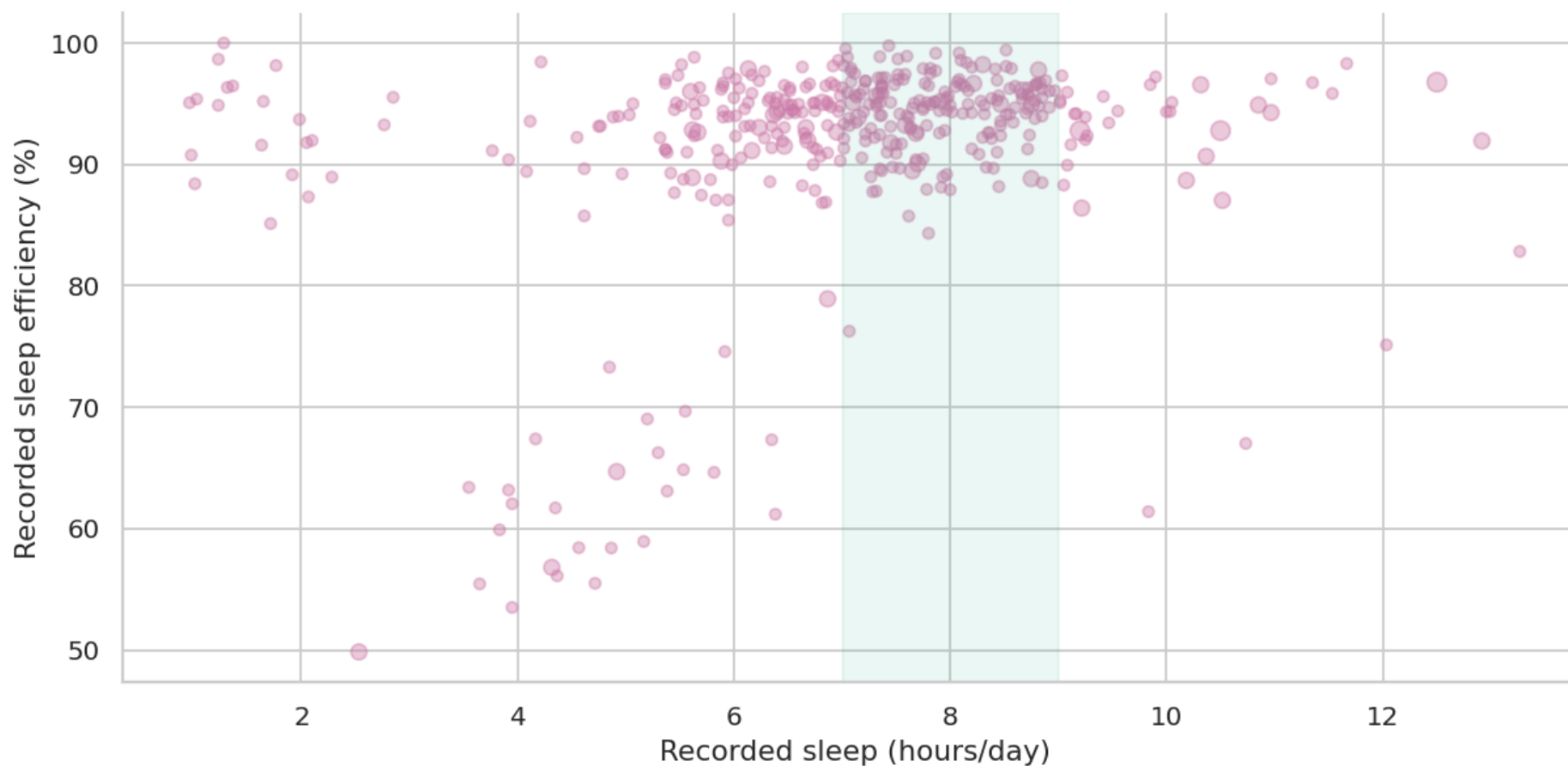
So what? Mean recorded sleep is 419.2 minutes (6.99 hours); 44.1% of days are below seven hours. The 7–9-hour band is an analytical reference, not an individualized prescription.

Sleep needs depend on age, which is absent. CDC recommends seven or more hours for adults 18–60, 7–9 hours for ages 61–64 and 7–8 hours for 65+. Therefore this notebook does not label all 9+ hour records unhealthy. [CDC sleep guidance](#).

Sleep-onset latency is not identifiable here. The in-bed minus asleep gap combines wakefulness before, during and after sleep; it is not time to fall asleep. `minuteSleep` has episode IDs and state codes, but no verified lights-out time and no attached state-code dictionary. All 459 deduplicated episodes are profiled without inventing a latency estimate.

```
plot_chart(19, data)
```

19 Time in bed exceeds sleep by 39 minutes on average



Source: supplied Fitbit / MTurk export, 2016. Descriptive observational analysis; see notebook denominators.

So what? Mean record-level efficiency is 91.6% and the mean in-bed gap is 39.3 minutes. Use this as a logging discussion, not a sleep-disorder diagnosis.

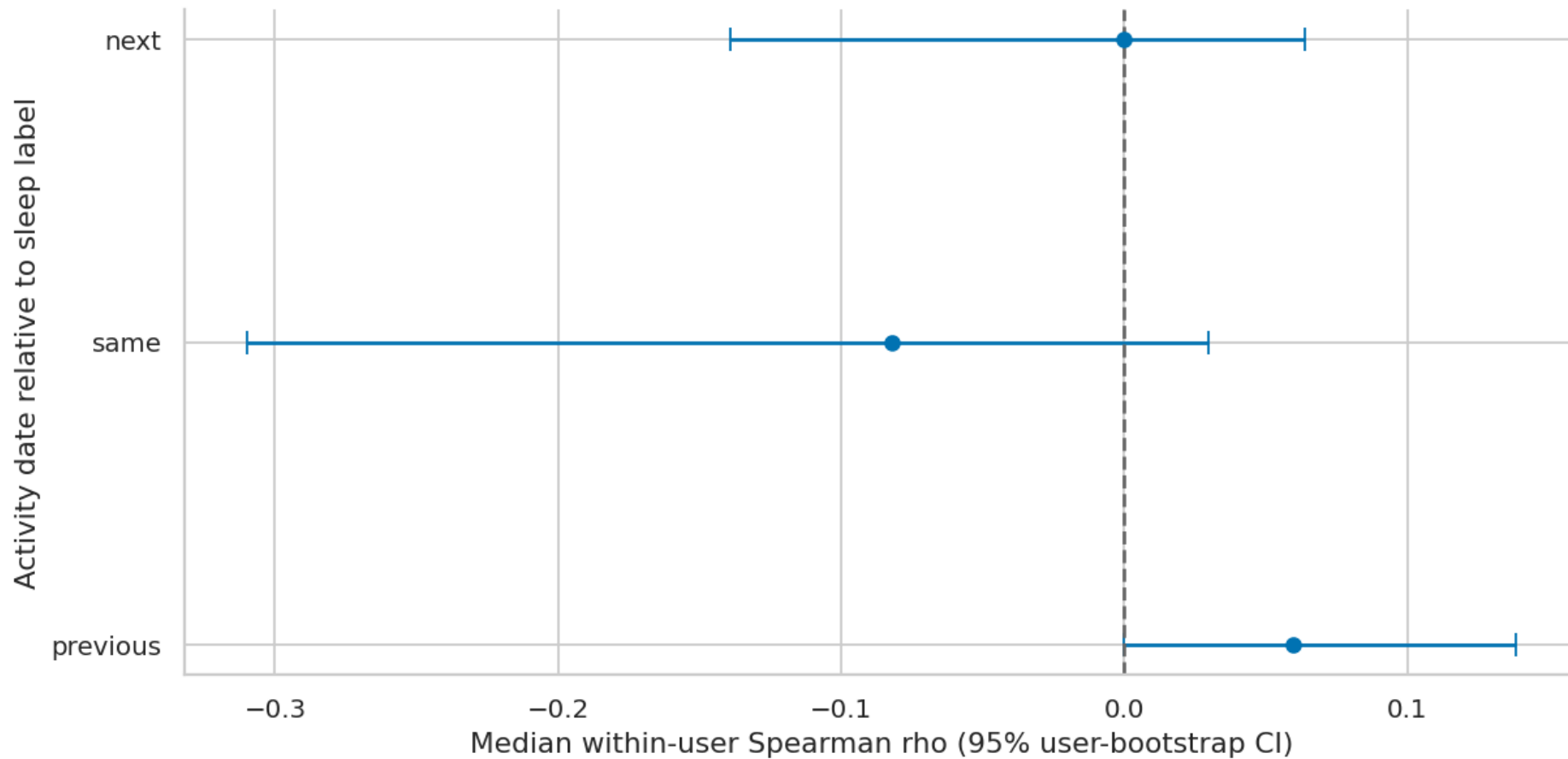
```
display(data["lag_stats"])
display(data["episodes"].head(8))
```

	alignmen t	users	pairs	median_ within_u ser_rho	ci_low	ci_high
0	previous	17	393	0.059628	0.000000	0.138340
1	same	18	408	-0.08194 0	-0.30978 9	0.029800
2	next	17	385	-0.00030 5	-0.13897 4	0.063774

	Id	logId	start_time	end_time	recorded_min utes	code_1_minut es	code_2_minut es	code_3_minut es
0	1503960366	11380564589	2016-04-12 02:47:30	2016-04-12 08:32:30	346	327	13	6
1	1503960366	11388770715	2016-04-13 03:08:30	2016-04-13 08:20:30	313	290	11	12
2	1503960366	11388770716	2016-04-13 20:10:00	2016-04-13 21:43:00	94	94	0	0
3	1503960366	11402722600	2016-04-15 02:59:00	2016-04-15 10:20:00	442	412	22	8
4	1503960366	11421831252	2016-04-16 02:11:00	2016-04-16 06:59:00	289	270	16	3
5	1503960366	11421831253	2016-04-16 07:02:00	2016-04-16 08:19:00	78	70	3	5
6	1503960366	11421831254	2016-04-16 23:27:00	2016-04-17 11:18:00	712	700	12	0
7	1503960366	11439580762	2016-04-19 02:06:30	2016-04-19 07:25:30	320	304	16	0

```
plot_chart(20, data)
```

20 Adjacent-day sleep associations are uncertain within this small panel



Source: supplied Fitbit / MTurk export, 2016. Descriptive observational analysis; see notebook denominators.

So what? Sleep is aligned to the export SleepDay date. Previous/same/next activity means exact calendar dates, with no forward filling. User-level correlation medians and user-bootstrap intervals avoid treating each day as an independent participant.

8. Heart-rate and intensity analysis

Takeaway: Only 14 users have heart-rate data; observed intervals are not equivalent to all-day exposure.

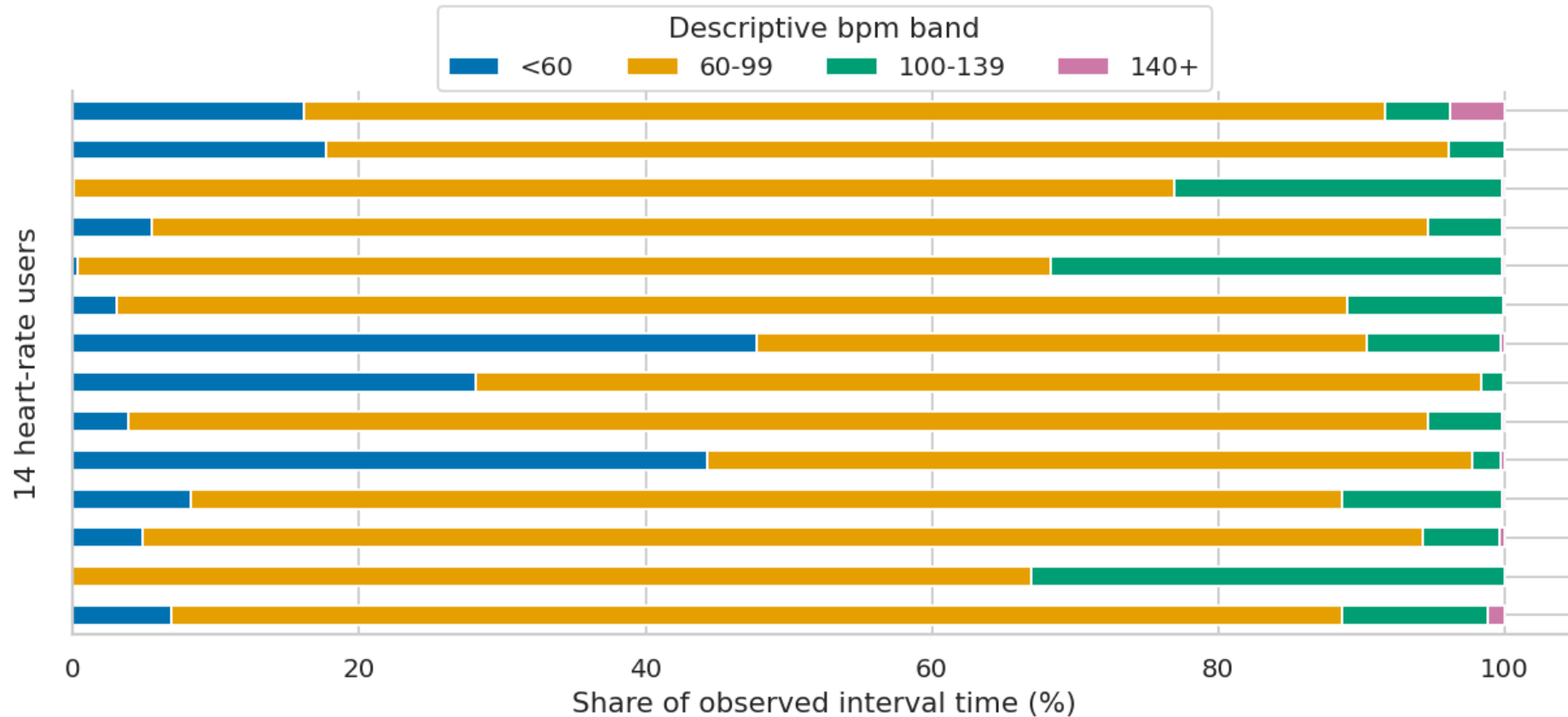
```
display(pd.read_csv(BASE / "sql_outputs/heart_rate_zones.csv"))
display(data["resting"])
```

	bpm_band	observed_hours	readings	excluded_gaps
0	100-139	395.917778	253915	146
1	140+	29.593611	35092	6
2	60-99	4114.031111	1821636	1049
3	<60	1002.786389	373015	123

	Id	night_p1_0_bpm
0	2026352035	67.198630
1	2347167796	61.088763
2	4020332650	58.683855
3	4388161847	47.966836
4	4558609924	61.966810
5	5553957443	56.670000
6	5577150313	48.161194
7	6117666160	61.883084
8	6775888955	67.418266
9	6962181067	59.813838
10	7007744171	76.200000
11	8792009665	58.008605
12	8877689391	70.414935

```
plot_chart(21, data)
```

21 Heart-rate time shares exclude gaps longer than 60 seconds

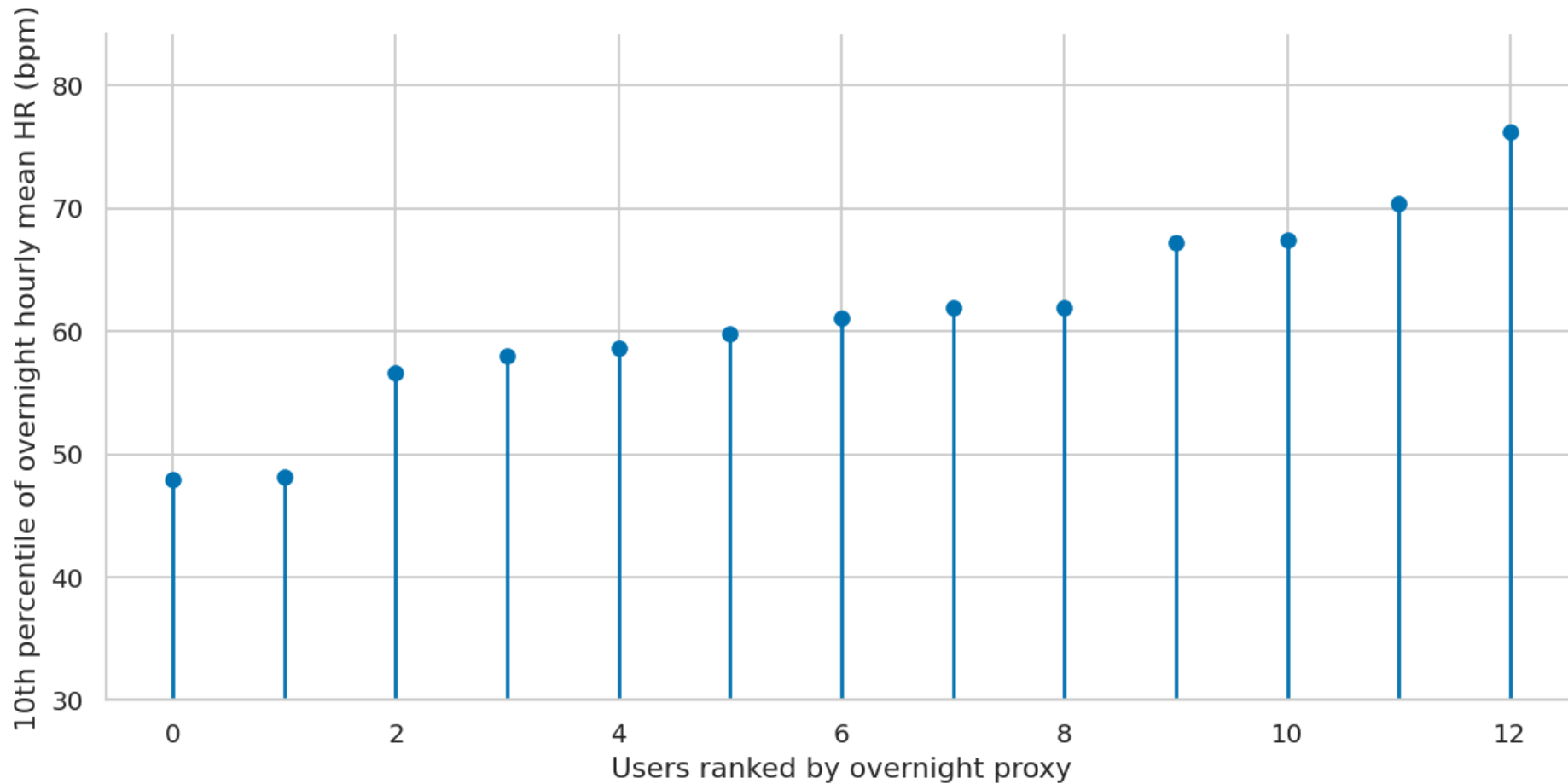


Source: supplied Fitbit / MTurk export, 2016. Descriptive observational analysis; see notebook denominators.

So what? Bpm bands are fixed descriptive ranges (<60, 60–99, 100–139, 140+), not personalized training zones. Each reading carries its next observed interval only if that interval is positive and ≤60 seconds; gaps and terminal readings add no duration.

```
plot_chart(22, data)
```

22 Overnight heart-rate proxies describe only users with night observations

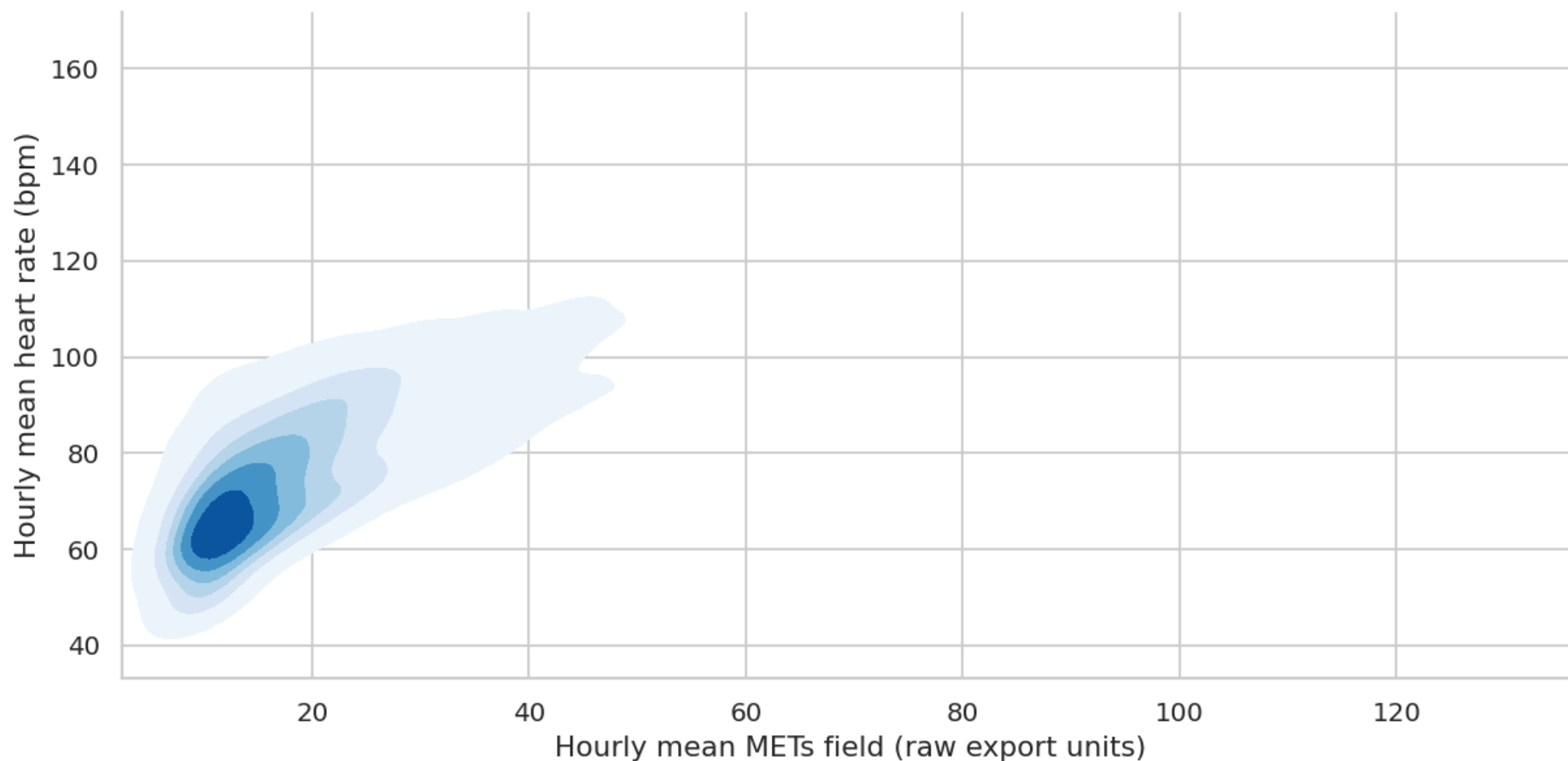


Source: supplied Fitbit / MTurk export, 2016. Descriptive observational analysis; see notebook denominators.

So what? The overnight proxy is the 10th percentile of hourly mean bpm between 00:00 and 05:59, calculated separately by user. It is neither clinical resting heart rate nor a sleep-stage measurement; sparse night coverage can bias it.

```
plot_chart(23, data)
```

23 5,990 matching user-hours link heart rate to raw exertion scores



Source: supplied Fitbit / MTurk export, 2016. Descriptive observational analysis; see notebook denominators.

So what? Only matching user-hours on retained activity dates enter this view. METs remains in raw export units. Repeated hourly observations and device algorithm overlap prevent an independent validation claim.

```
hr_user = data["hr_join"].groupby("Id")[["mean_bpm", "TotalIntensity", "mean_raw_mets"]].mean()
display(hr_user.corr(method="spearman").round(3))
print(f"HR/intensity/MET comparison: {len(data['hr_join']):,} hours, {len(hr_user)} users")
```

HR/intensity/MET comparison: 5,990 hours, 14 users

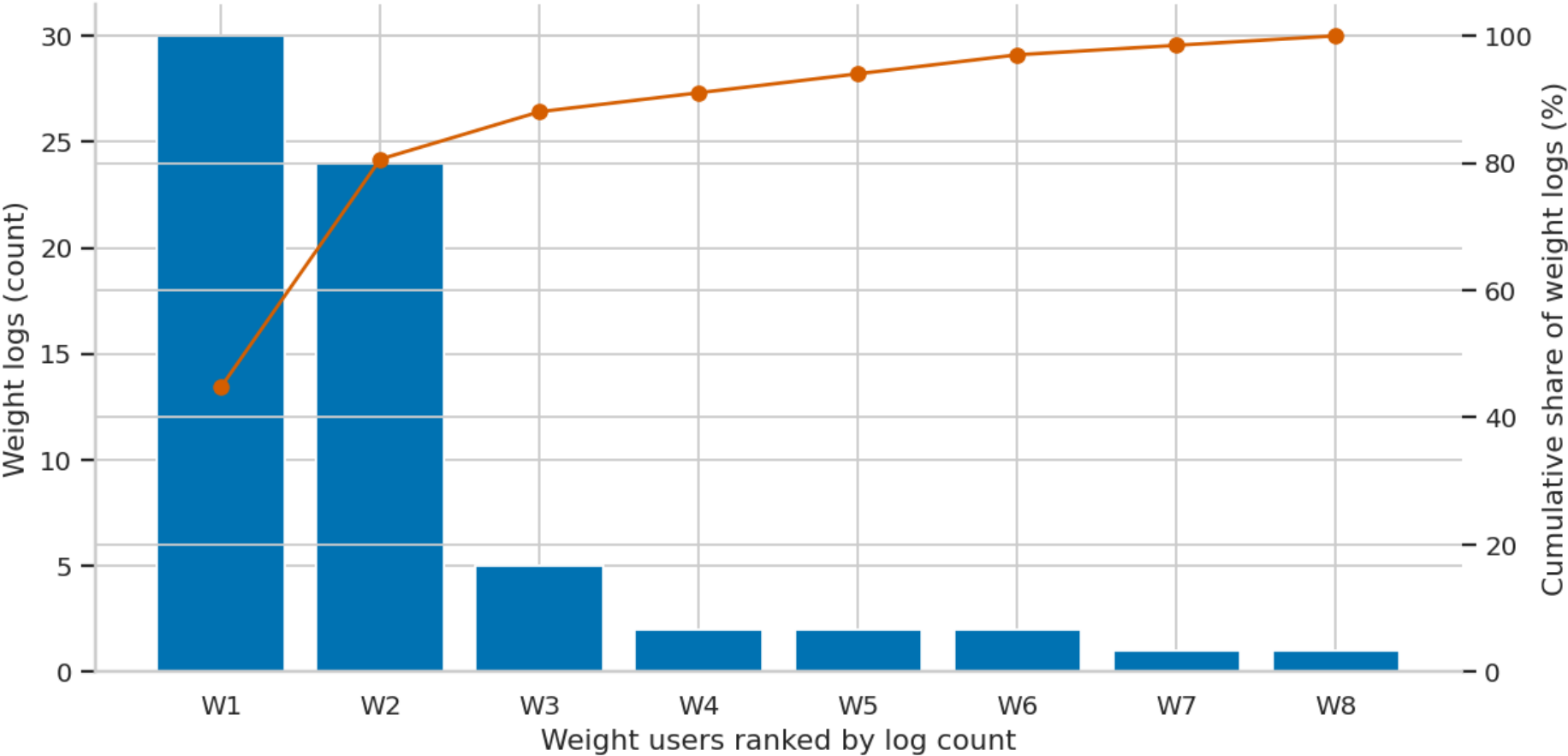
	mean_bpm	TotalIntensity	mean_raw_mets
mean_bpm	1.000	0.481	0.477
TotalIntensity	0.481	1.000	0.947
mean_raw_mets	0.477	0.947	1.000

9. Weight/BMI logging behaviour

Takeaway: The weight sample supports a feature-engagement observation, not a weight-management conclusion.

```
plot_chart(24, data)
```

24 Top two weight loggers contribute 80.6% of records

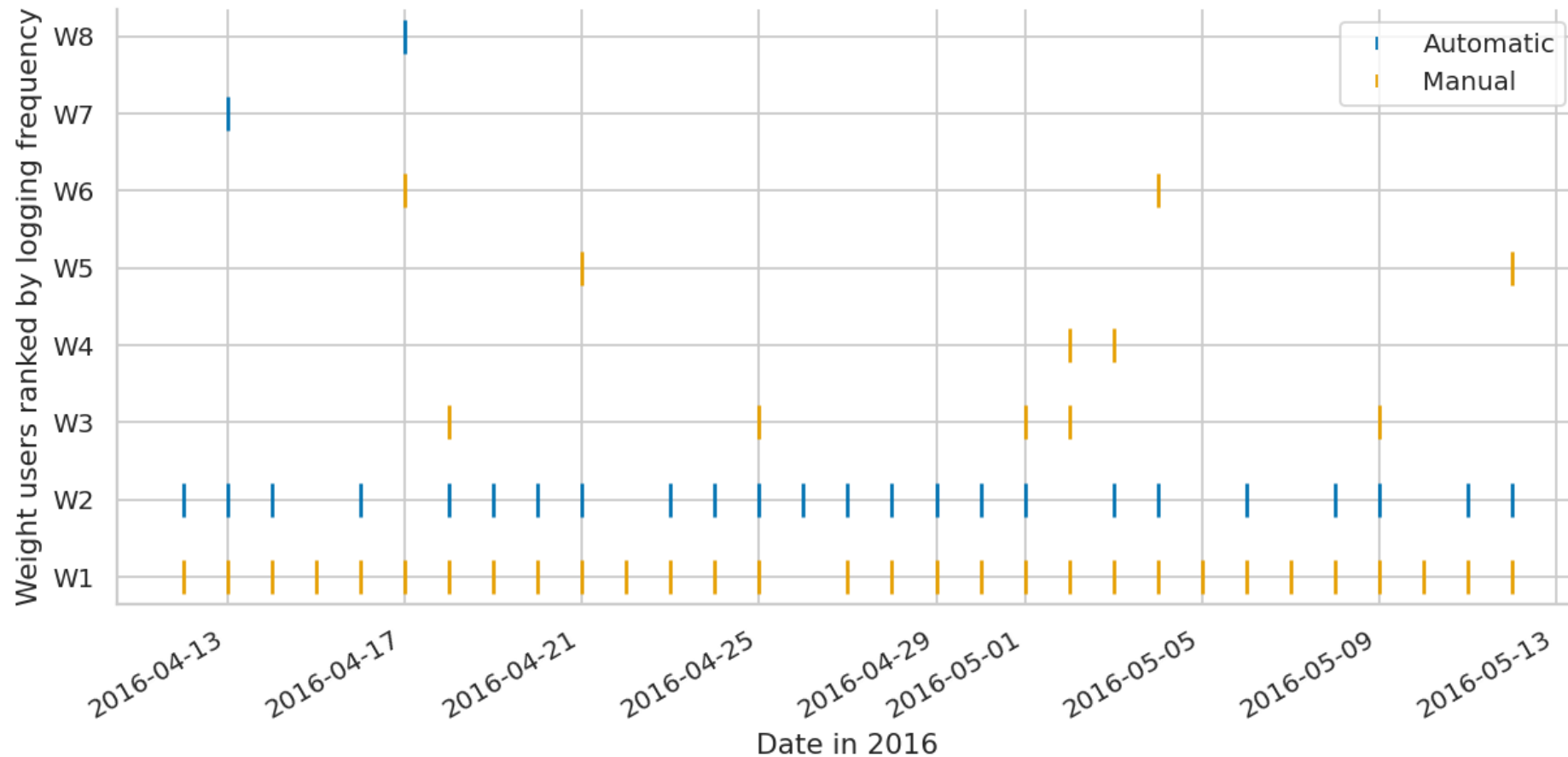


Source: supplied Fitbit / MTurk export, 2016. Descriptive observational analysis; see notebook denominators.

So what? Highly concentrated logging makes record-weighted weight/BMI averages unrepresentative even of the eight loggers. Acquisition targeting based on body size would be unsupported.

```
plot_chart(25, data)
```

25 41 of 67 weight logs are manual



Source: supplied Fitbit / MTurk export, 2016. Descriptive observational analysis; see notebook denominators.

So what? 41/67 logs (61.2%) are manually reported. Investigate logging friction with consented research; the data cannot explain why other users have no weight logs.

```
display(pd.read_csv(BASE / "sql_outputs/weight_behavior.csv"))
```

	Id	logs	days	manual_logs	missing_fat	mean_recorded_bmi
0	1503960366	2	2	2	1	22.650000
1	1927972279	1	1	0	1	47.540001
2	2873212765	2	2	2	2	21.570001
3	4319703577	2	2	2	1	27.415000
4	4558609924	5	5	5	5	27.214000
5	5577150313	1	1	0	1	28.000000
6	6962181067	30	30	30	30	24.028000
7	8877689391	24	24	0	24	25.487083

10. User segmentation

Takeaway: Interpretable rules guide a first test; KMeans is an exploratory cross-check on 33 users.

Rule-based segments, in priority order: Reconnect = fewer than 21 retained days; Build routine = at least 21 retained days and mean steps below 7,500; Keep momentum = remaining users. Thresholds are operational choices, not validated medical categories.

KMeans: one row per user; standardized mean steps, mean active-minute share and engagement score. Sleep and weight are excluded to avoid imputing sparse modalities; demographic features are unavailable. Candidate k=2...6 uses 30 initializations and fixed seed 42. Choose the highest silhouette, inspect the elbow and report the actual profile. Naming describes recorded behaviour and does not infer motivation or demographic identity.

```
display(u.groupby("rule_segment").agg(users=("Id", "size"), steps=("mean_steps", "mean"), engagement=("engagement_score", "mean")))
display(data["scores"])
display(data["profile"])
```

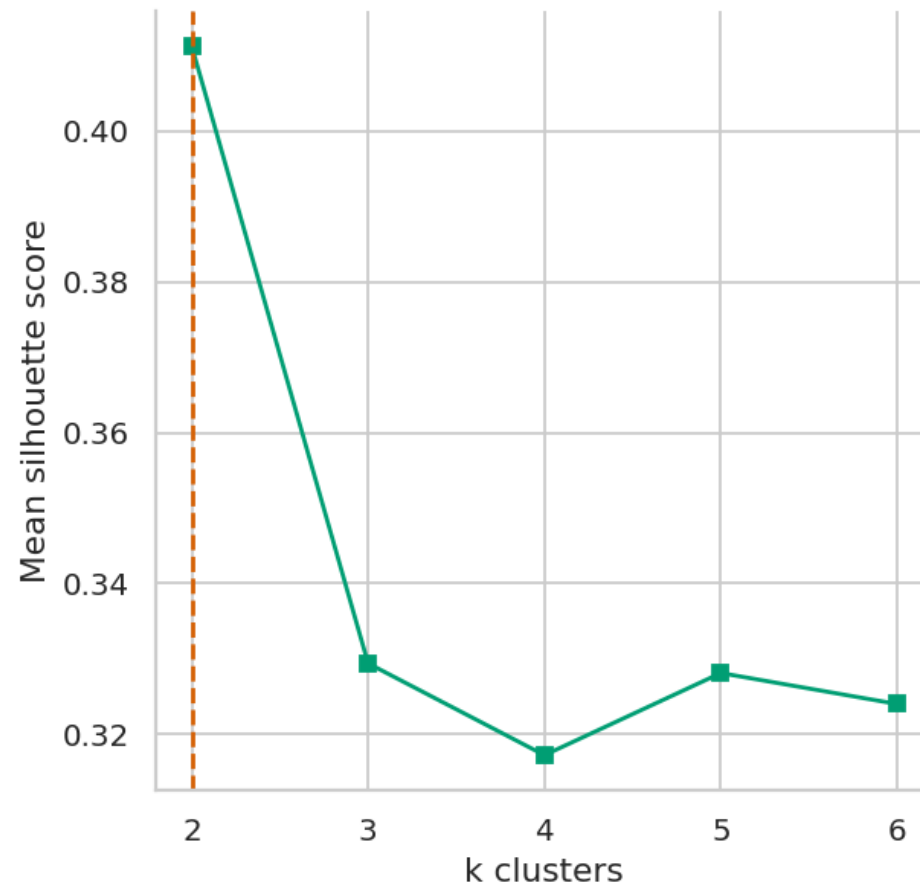
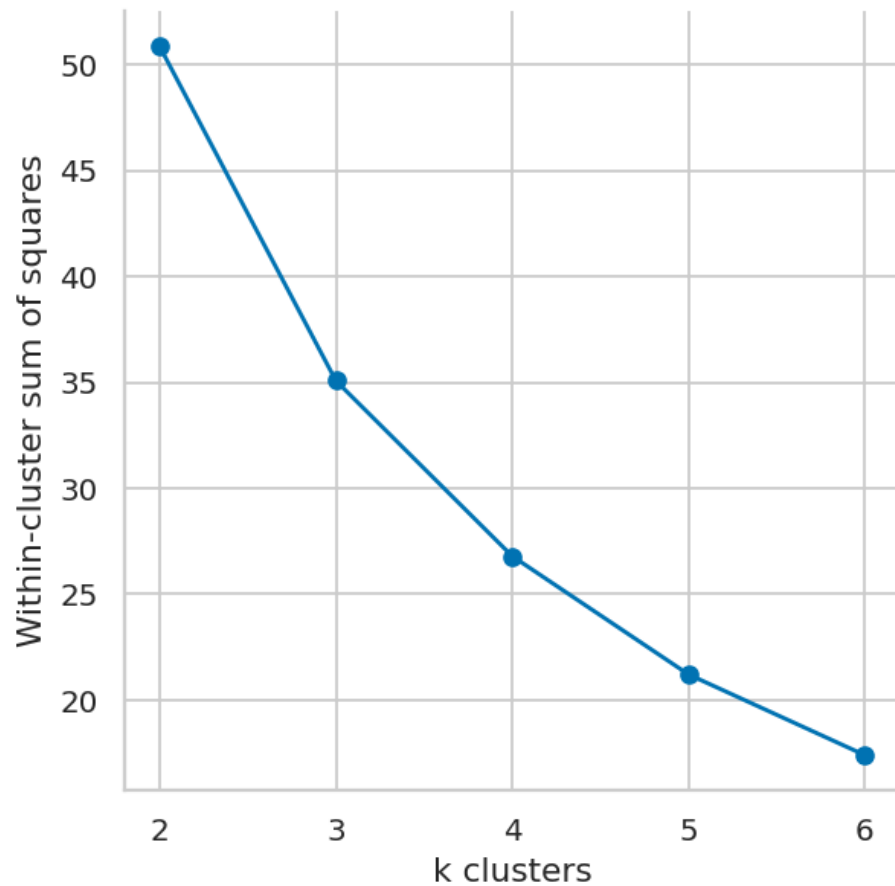
rule_segment	users	steps	engagement
Build routine	6	5822.283647	95.698925
Keep momentum	18	10161.087048	94.086022
Reconnect	9	5179.229452	51.254480

	k	inerti a	silho uette
0	2	50.88 0982	0.411 270
1	3	35.08 2541	0.329 410
2	4	26.79 8397	0.317 233
3	5	21.21 1531	0.328 128
4	6	17.40 0303	0.324 016

	cluster	mean_step s	mean_acti ve_share	engageme nt_score	persona	users
0	0	5287.41316 4	0.141009	66.253102	Gentle starters	13
1	1	9785.49813 5	0.257818	93.387097	Routine builders	20

```
plot_chart(26, data)
```

26 k=2 maximizes silhouette at 0.41

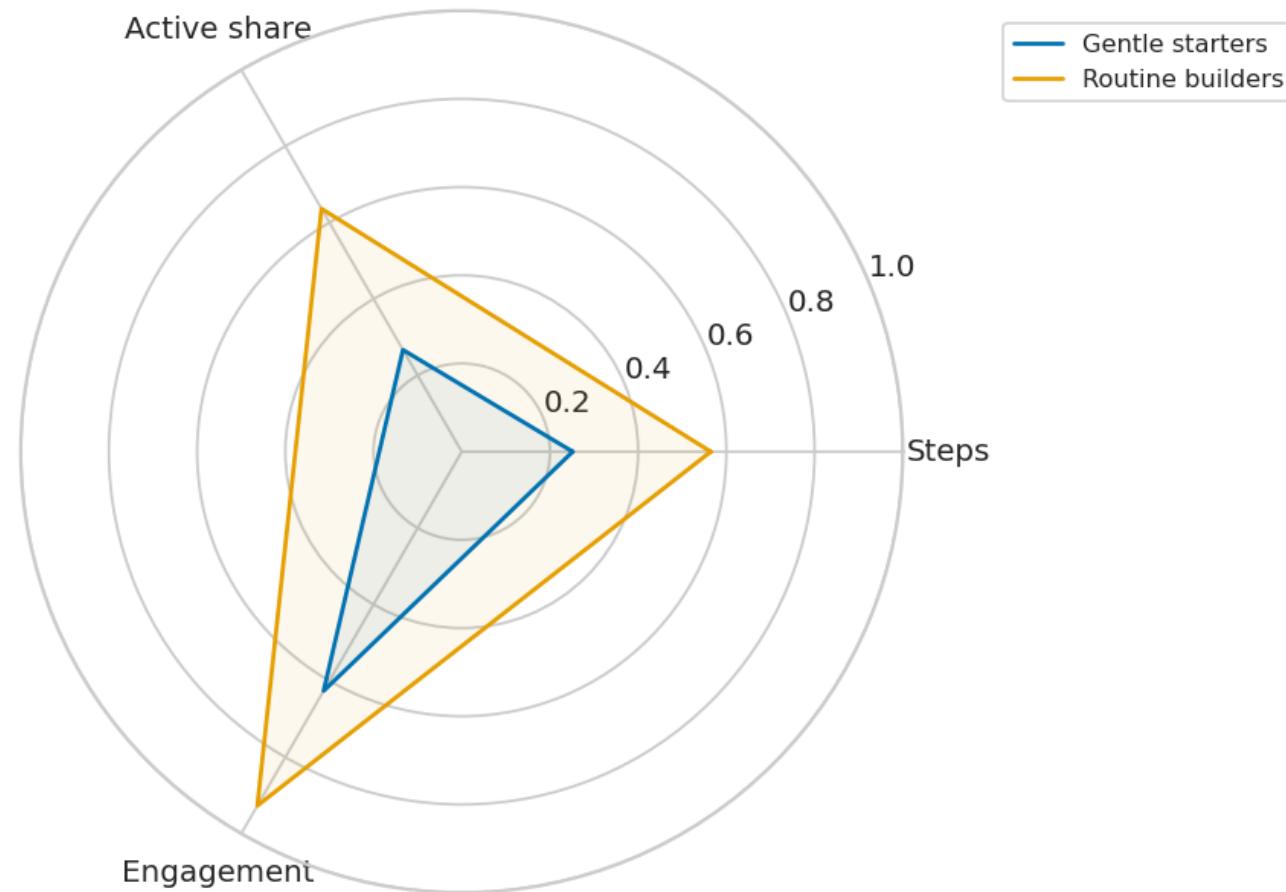


Source: supplied Fitbit / MTurk export, 2016. Descriptive observational analysis; see notebook denominators.

So what? k=2 has silhouette 0.411; the elbow is shown as a secondary diagnostic. Ten alternate initialization seeds produce minimum adjusted Rand index 1.00. This tests initialization stability, not population or resampling stability.

```
plot_chart(27, data)
```

27 Personas differ in activity and recording consistency (min-max profile)

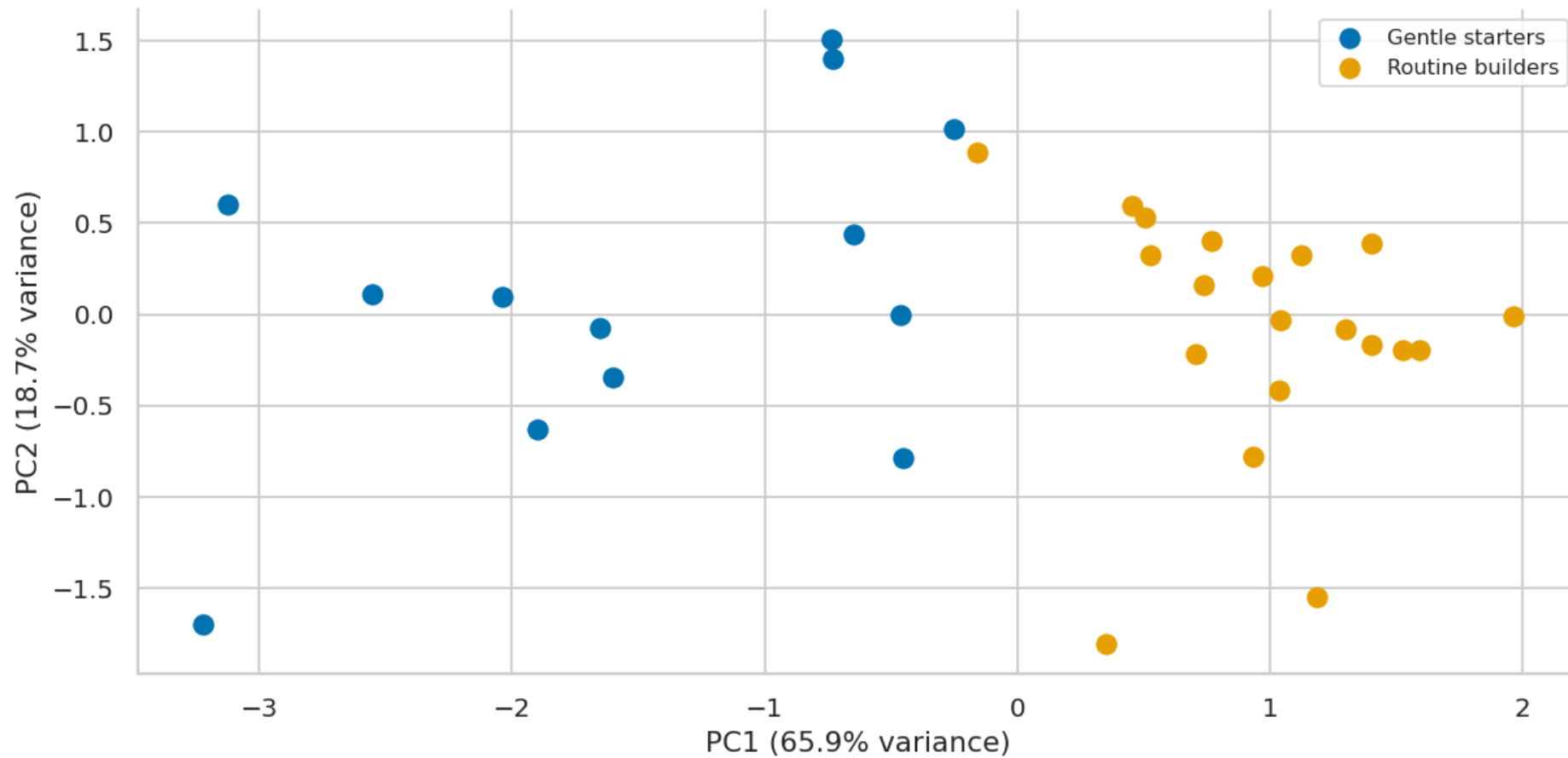


Source: supplied Fitbit / MTurk export, 2016. Descriptive observational analysis; see notebook denominators.

So what? Each axis rescales the feature to the observed user minimum and maximum; polygons do not represent a composite wellness score. Use the adjacent profile table for the original values and persona counts.

```
plot_chart(28, data)
```

28 Clustering describes this sample; the personas need external validation

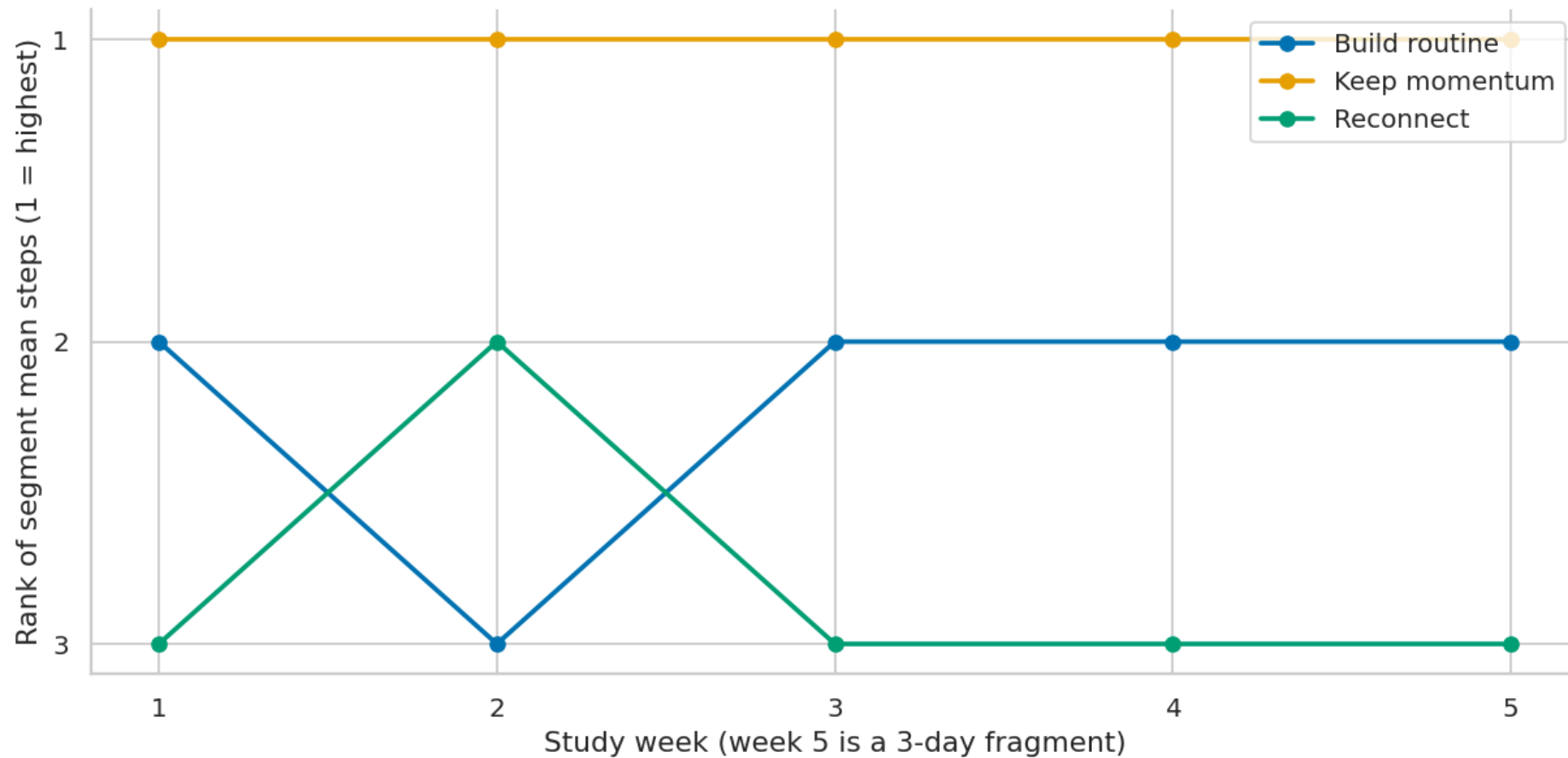


Source: supplied Fitbit / MTurk export, 2016. Descriptive observational analysis; see notebook denominators.

So what? PCA only visualizes the standardized feature space. Clusters are sample descriptions; test future cohorts and resampling stability before operational assignment.

```
plot_chart(30, data)
```

30 Rule-based segments provide a stable, interpretable campaign vocabulary



Source: supplied Fitbit / MTurk export, 2016. Descriptive observational analysis; see notebook denominators.

So what? Rule segments are assigned from the full month, so their weekly separation is descriptive and partly built into the definition. A prospective campaign would assign segments using a prior observation period.

11. Statistical validation

Takeaway: Small user counts and repeated measures limit inferential strength; effect sizes and uncertainty take priority.

```
display(data["tests"].round(5))
```

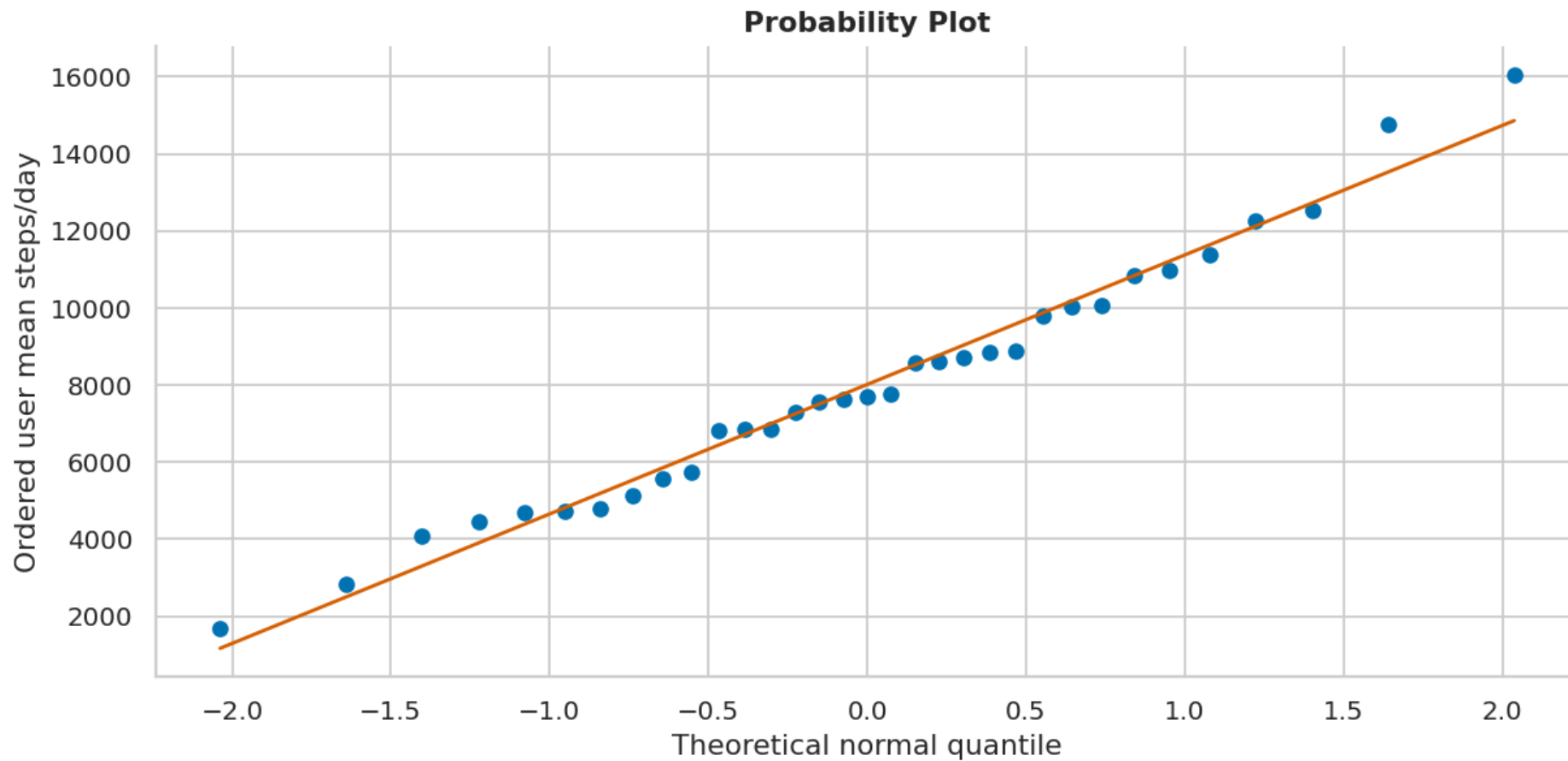
	test	comparison	n	statistic	p_value	effect_size	effect_definition	p_holm
0	Shapiro-Wilk	mean_steps	33	0.98170	0.83597	NaN	W reported in statistic	NaN
1	Shapiro-Wilk	mean_active	33	0.96995	0.47884	NaN	W reported in statistic	NaN
2	Shapiro-Wilk	mean_calories	33	0.94995	0.13256	NaN	W reported in statistic	NaN
3	Spearman	mean_steps / mean_calories	33	0.31283	0.07630	0.31283	Spearman rho; one mean per user	0.38148
4	Spearman	mean_steps / mean_sleep	24	-0.13043	0.54353	-0.13043	Spearman rho; one mean per user	1.00000
5	Spearman	mean_active / mean_sleep	24	0.07130	0.74057	0.07130	Spearman rho; one mean per user	1.00000
6	Mann-Whitney U	Sleep loggers vs nonloggers: mean steps	33	114.00000	0.82406	0.05556	Cliff delta; positive favors sleep loggers	1.00000
7	Wilcoxon paired	Weekend minus weekday user mean steps	32	245.00000	0.73288	0.07197	Matched rank-biserial; positive favors weekend	1.00000

All between-user tests use one summary per user. Shapiro–Wilk checks are diagnostics, not proof of normality. Spearman tests assess monotonic associations. Mann–Whitney U compares the 24 sleep loggers against nine other users, with Cliff’s delta as effect size. Wilcoxon pairs the 32 users with both weekday and weekend data, reporting rank-biserial effect size. Holm correction covers the five association/group hypotheses, while three normality diagnostics are separate.

The weekend confidence interval resamples users 2,000 times. Sleep lag confidence intervals resample eligible user correlations, requiring at least five complete pairs per user. These are exploratory estimates, not randomized treatment effects. No significance tests are run on clusters defined using the same activity features, which would be circular.

```
plot_chart(29, data)
```

29 Normality diagnostics use 33 independent user summaries



Source: supplied Fitbit / MTurk export, 2016. Descriptive observational analysis; see notebook denominators.

So what? The user-level Shapiro tests do not reject normality at 0.05; that does not establish a normal population. Nonparametric tests retain a consistent approach for this small, selected panel.

Inference summary: none of the five association/group hypotheses survives the Holm-adjusted 0.05 threshold. This is not evidence that relationships do not exist; the sample is small, self-selected and lacks confounders. Daily and hourly charts remain descriptive. Marketing proposals are experiments justified by observed usage, not claimed effects on health or retention.

12. Key findings

Takeaway: The strongest evidence concerns recording quality, timing and uneven feature use.

1. **Coverage:** 33 activity users, 24 sleep users, 14 heart-rate users and eight weight users (Chart 01).
2. **Wear-quality sensitivity:** 94/940 activity days excluded; mean steps move from 7,638 to 8,427 under the proxy (Chart 03).
3. **Movement baseline:** 45.4% of retained days stay below 7,500 steps (Chart 04).
4. **Light activity dominates:** 84.7% of classified active minutes are light (Chart 16).
5. **Timing:** 18:00 is the pooled peak, 660 steps per observed hour (Chart 14).
6. **Sleep:** mean 419.2 minutes; 44.1% of unique sleep days below seven hours (Chart 18).
7. **Weekend uncertainty:** paired gap +120 steps/day, with 95% CI spanning -959 to +1143 (Chart 13).
8. **Weight friction hypothesis:** 41/67 logs manual and 65/67 missing Fat; eight users provide all weight records (Charts 02, 25).

13. Marketing recommendations: Bellabeat App/Membership

Takeaway: Test activity-led onboarding and optional sleep value before expanding a weight-led proposition.

Recommendation	Evidence	Action	KPI
Quality-aware onboarding	94/940 days excluded; raw/clean mean differs by 789 steps	Test opt-in setup/sync checks and explain missing data before issuing activity nudges	Retained recording days / eligible days; seven-day activation; notification opt-out
Personal baseline campaigns	45.4% of retained days <7,500 steps; light activity is 84.7% of active minutes	Test incremental movement goals matched to prior activity and the rule personas	Change in user-level retained-day steps; goal engagement; adverse/negative feedback
Timing experiment	18:00 peak averages 660 steps per observed hour	Randomize individualized reminder timing against a fixed schedule and a no-reminder holdout	Incremental weekly active days; prompt engagement; notification fatigue
Optional sleep onboarding	24/33 sleep loggers; 44.1% of sleep days <7 h	Test a sleep-summary onboarding path and consistent-routine content for consenting users	Sleep-feature activation; 14-day logging continuity; membership conversion
Keep weight optional	8/33 weight loggers; 41/67 entries manual	Prioritize activity/sleep value messages; interview weight loggers before adding weight-led campaigns	Feature opt-in and completion; entry burden; D30 retention by randomized arm

Measurement design: recruit a contemporary, consented Bellabeat target-market cohort; assign experiments at user level, using prior-period segments. Predefine one primary KPI and guardrails, minimum detectable effect and power before sample sizing. Track missingness as an outcome; improved retention can change the observed activity mix. Randomize within baseline segment, use intention-to-treat reporting and report confidence intervals. Subscription conversion and true app retention are **future instrumentation requirements**; neither can be estimated from the supplied files. No numeric uplift forecast is defensible here.

14. Limitations, bias and next steps

Takeaway: This ten-year-old convenience sample generates testable hypotheses, not a representative market forecast.

- **Population:** 33 self-selected MTurk activity users, with no age, sex, height, profession, geography or confirmed membership in Bellabeat's audience. No women-specific inference is made.
- **Time:** one 31-day interval in 2016; the last day and several user histories are partial. No seasonal or long-term habit claim is supportable.
- **Missingness:** absent sleep/weight/HR may reflect devices, permissions or export coverage rather than preference. Nonwear cannot be confirmed. Conditioning on the quality proxy can bias activity upward; thresholds are sensitivity-tested.
- **Measurement:** proprietary classifications and calorie estimates; raw distance/intensity/MET units not verified. No calorie intake, stress, menstrual-cycle or health-outcome evidence exists.
- **Sleep:** daily labels and multiple episodes limit nightly interpretation; no reliable sleep-onset latency. Adjacent-day alignment is an explicit calendar convention.
- **Statistics:** repeated measurements, sparse weight logs, modest independent user counts and multiple exploratory questions. No causal effects; cluster initialization stability is not external validity.
- **Commercial evidence:** no acquisition, subscription, notification exposure, retention instrumentation or revenue data.

Next steps: validate a source dictionary and timezone metadata; collect on-body wear signals and enrollment dates; instrument opt-in app events; recruit a representative target cohort; pre-register randomized onboarding/reminder tests; externally validate personas. The processed tables are prepared for later Power BI and Streamlit work, which is intentionally deferred.

15. Appendix: reproducibility and reconciliation

Takeaway: Python and SQL agree on every selected parity metric and rule-segment count.

```
display(data["recon"])
assert data["recon"]["match"].all()
print("All reconciliation checks passed.")
print(f"Charts saved: {len(list((BASE / 'charts').glob('chart_*.png')))}")
```

```
All reconciliation checks passed.
Charts saved: 30
```

	metric	python	sql	match
0	activity_uses	33.000000	33.000000	True
1	clean_activity_rows	846.000000	846.000000	True
2	average_steps	8426.832151	8426.832151	True
3	clean_sleep_rows	410.000000	410.000000	True
4	average_sleep_minutes	419.173171	419.173171	True
5	clean_weight_rows	67.000000	67.000000	True
6	segment: Keep momentum	18.000000	18.000000	True
7	segment: Reconnect	9.000000	9.000000	True
8	segment: Build routine	6.000000	6.000000	True

Rebuild order: create a Python environment from `requirements.txt` ; import the 18 CSVs with `load_csv_to_sqlite.py` ; execute `run_sql.py` ; run this notebook top to bottom.

`run_all.py --raw-dir ../raw` automates import, SQL, notebook execution and HTML/PDF export. SQL stores ISO dates as TEXT; date normalization happens explicitly at import, never via SQLite's ambiguous interpretation of US dates.

Traceability: `source_inventory` records file hashes, rows, columns, users and bounds; `column_audit` records every field; `cleaning_log` records transformations; `sql_outputs/execution_log.csv` lists every executed master-SQL statement; `reconciliation` compares independently cleaned pandas data to SQL. CSV and parquet pairs share each processed table's rows. IDs remain 64-bit integers; treat them as identifiers in downstream tools. Date fields and joins are documented in `processed/DATA_DICTIONARY.md`.

Chart register: 30 distinct numbered views. Chart 01 modality adoption bars; 02 missing-cell audit; 03 threshold sensitivity; 04 histogram/KDE; 05 ECDF; 06 box plots; 07 violins; 08 sleep-record strip plot; 09 correlation matrix; 10 hexbin; 11 regression/CI; 12 lollipop; 13 paired dumbbell; 14 hour×weekday heatmap; 15 calendar heatmap; 16 stacked area; 17 trend/coverage panels; 18 sleep composition; 19 efficiency bubble plot; 20 lag-association intervals; 21 observed HR-time composition; 22 overnight proxy stems; 23 bivariate KDE contours; 24 Pareto; 25 event timeline; 26 elbow/silhouette diagnostics; 27 radar; 28 PCA cluster projection; 29 Q–Q; 30 segment-rank bump chart. Reused visual primitives answer different questions at different grains; no chart is a cosmetic recolouring of another.

External references: [WHO activity guidance](#), [CDC sleep guidance](#). Step bands follow the task's requested historical descriptive convention, not a CDC universal 10,000-step requirement. Product strategy is scoped to the supplied case brief rather than asserted current Bellabeat features.

Execution environment: all cells were executed sequentially with IPython and real outputs captured because this hosted environment disallows kernel sockets. `execute_notebook.py` supplies that reproducible path; ordinary Jupyter/nbclient execution remains supported. No error outputs were suppressed or fabricated.

```
import platform
from importlib.metadata import version
packages = ["pandas", "numpy", "scipy", "scikit-learn", "matplotlib", "seaborn", "pyarrow", "nbformat", "nbclient", "nbconvert", "weasyprint"]
versions = {name: version(name) for name in packages}
versions["python"] = platform.python_version()
```

```
versions["sqlite"] = sqlite3.sqlite_version
pd.Series(versions, name="version").to_csv(BASE / "processed/software_versions.csv")
display(pd.Series(versions, name="version"))
```

```
pandas      2.2.3
numpy        2.3.5
scipy        1.17.0
scikit-learn 1.8.0
matplotlib  3.10.8
seaborn      0.13.2
pyarrow      25.0.1
nbformat     5.11.1
nbclient     0.11.0
nbconvert    7.17.1
weasyprint   70.0
python       3.12.14
sqlite       3.53.1
Name: version, dtype: object
```